MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION DOCUMENT
49EE SEWAGE TREATMENT PLANT
DESIGN OF SURGE SEWAGE & SUMP TANK

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RC	:	Reinforced Concrete

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1. INTENT

This document is to Prepare for design of Surge Sewage Tank & Sump Tank of **Sewage Treatment Plant**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the sump tank

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For all design strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Base Pressure check are shown in STAAD Pro. (for mat footing). For Base slab, Wall and Top slab STAAD modelling has been done with Plate Elements. Plate mat with Vertical Subgrade Modulus of 102960 kN/m2/m has been considered (as per soil report BH-22).

4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are as per Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-0000-EG-CV-CRT-0001).

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4.2 Load Combination

Load combinations are based on SBC-301,2007, MD-216-1000-EG-ST-CRT-1001, ACI 318M

Basic Load Cases are: -

- D= Dead load
- L= Live Load
- F = Fluid Load F1=Fluid Load in Sump Tank Part A F2=Fluid Load in Sump Tank Part B
- H = Soil Load

Allowable Stress Design (Surge Sewage Tank)

Load Case	Load Combination
201	D + F
202	D + F + L + H
203	D + F + H

Ultimate Strength Design (Surge Sewage Tank)

Load Case	Load Combination
101	1.4 (D + F)
102	1.2 (D + F) + 1.6 (L+ H)
103	1.4 (D + H)

Allowable Stress Design (Sump Tank)

Load Case	Load Combination
201	D + F1
202	D + F2
203	D+F1+F2
204	1.0 (D + F1) + 1.0 (L+ H)
205	1.0 (D + F2) + 1.0 (L+ H)
206	1.0 (D +F1+ F2) + 1.0 (L+ H)
207	1.0 (D + H)

Ultimate Strength Design (Sump Tank)

Load Case	Load Combination
101	1.4 (D + F1)
102	1.4 (D + F2)
103	1.4 (D+F1+F2)
104	1.2 (D+F1) + 1.6 (L+ H)
105	1.2 (D+F2) + 1.6 (L+ H)
106	1.2 (D+F1+F2) + 1.6 (L+ H)
107	1.4 (D + H)

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5. CONCRETE SURGE SEWAGE TANK & SUMP TANK

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-301, SBC-303, SBC-304, ACI-318. The analysis has been carried out in Staad pro & designed by Excel Sheet. For all design strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Base Pressure check are shown in STAAD Pro. (for mat footing). For Base slab, Wall and Top slab STAAD modelling has been done with Plate Elements. The Vertical Modulus of Subgrade Reaction has been considered as 102960 kN/m^2/m at 1.5m depth. Refer chart BH-22 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

5.2 Materials

Concrete Strength of 28 days.

- Foundation and Structural Members: 35 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade:

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Contact with ground	Wall and Base Slab	50
Exposed to Air	Top Slab	25

5.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm. except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

5.5 Allowable Soil Bearing Capacity

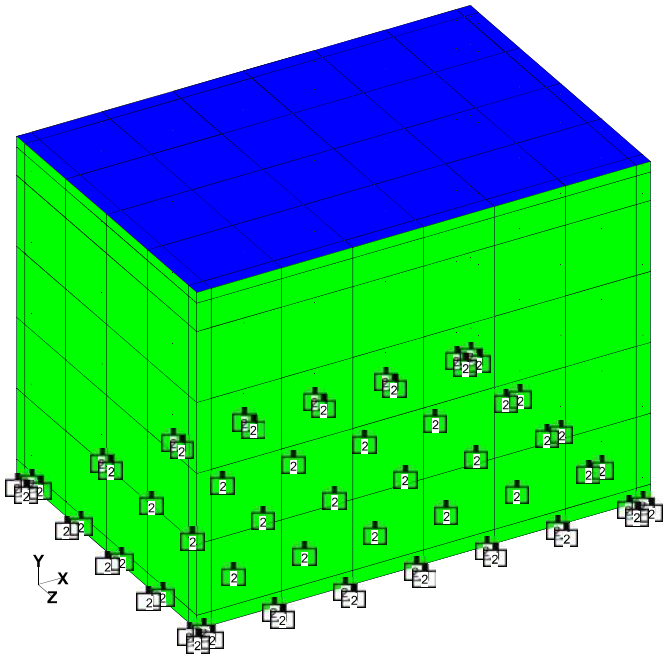
The net safe bearing capacity of soil for foundations design has been considered 1144 kN/ m^2 at 1.5 m below EGL refer chart location: BH-22 Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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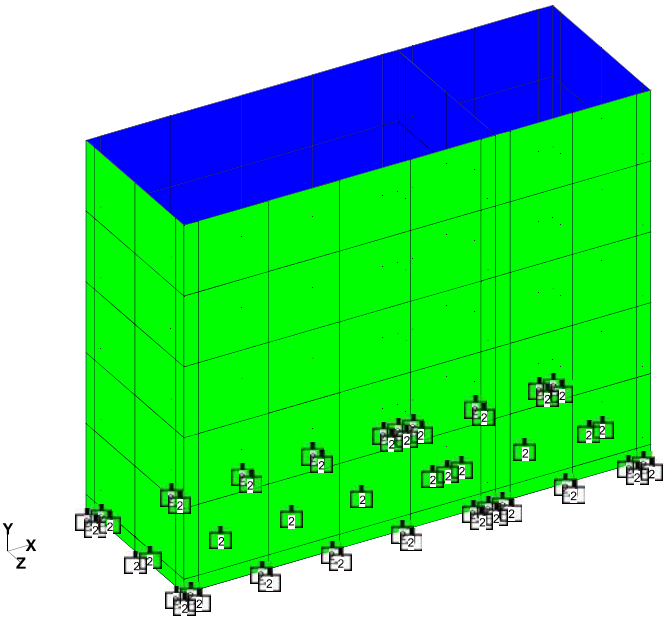
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6 STAAD GEOMETRY



SURGE SEWAGE TANK



SUMP TANK

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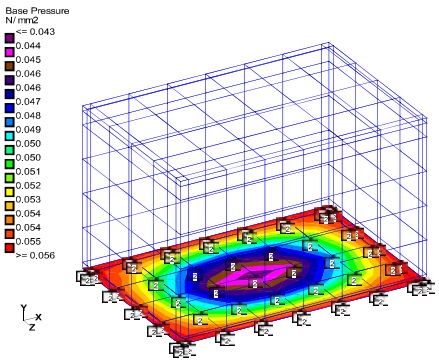
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7. DESIGN CALCULATIONS FOR SURGE SEWAGE TANK

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Design of Surge Sewage Tank Base Slab
(Refer staad file SURGE SEWAGE TANK.std)



For Calculation of Base pressure, Plate stress for wall, Slab Refer STAAD File

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +920.450m)
Thickness of Base Slab	D	=	0.20	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	1144	KN/m^2	
Gross Bearing Capacity		=	1171	KN/m^2	
Earth Pressure at rest Ko		=	0.5		Ref Soil report
Vertical modulus of Subgrage Reaction in staad		=	102960	KN/m^3	Staad input for Mat Fdn.
Water Pressure for 2.2 m height			22	KN/m^2	Applied in Staad
Soil pressure for 2.0m height			18	KN/m^2	Applied in Staad
50mm Screed pressure applied on base of slab			1.2	KN/m^2	Applied in Staad
Surcharge Pressure			10	KN/m^2	Applied in Staad

1) Check for Gross Base Pressure :

Maximum Base Pressure, Pmax	=	56.0	KN/m^2	Load Case 202
Minimum Base Pressure, Pmin	=	23.0	KN/m^2	Load Case 203
				Ok

2) Design of Base Slab :

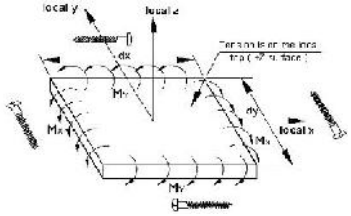
Slab Thickness =	200 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	3.050	kN-m	Corresponding Mxy	-0.226	kN-m	*(Plate No. - 1681 , L/C No. - 1.4*)*
Min. Mx	-0.797	kN-m	Corresponding Mxy	-0.656	kN-m	*(Plate No. - 1679 , L/C No. - 1.4*)*
Max. My	5.312	kN-m	Corresponding Mxy	-0.226	kN-m	*(Plate No. - 1681 , L/C No. - 1.4*)*
Min. My	-1.532	kN-m	Corresponding Mxy	-0.286	kN-m	*(Plate No. - 1675 , L/C No. - 1.4*)*

FOR TOP FACE REINFORCEMRNT :



Mx1 =	Mx + abs(Mxy) =	3.05 + 0.226 =	3.276	kN-m
My1 =	My + abs(Mxy) =	5.312 + 0.226 =	5.538	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	3.05 + 0.01 =	3.06	kN-m
My2 =	My+abs(Mxy^2/Mx) =	5.312 + 0.017 =	5.329	kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 3.276 kN-m
Myd = 5.538 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 3.276 kN-m
Φ*Kn = Mu/bd^2 deff. = 144 mm
Mu/bd^2 = 3.276X10^6/(1000X144^2) = 0.158 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X144/100 = 0.58 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 5.538 kN-m
Φ*Kn = Mu/bd^2 deff. = 132 mm
Mu/bd^2 = 5.538X10^6/(1000X132^2) = 0.267 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X144/100 = 0.92 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-0.797 - 0.656 =	-1.453 kN-m
My1 =	My - abs(Mxy) =	-1.532 - 0.286 =	-1.818 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-0.797 - 0.281 =	-1.078 kN-m
My2 =	My-abs(Mxy^2/Mx) =	-1.532 - 0.103 =	-1.429 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -1.453 kN-m
Myd = -1.818 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 1.453 kN-m
Φ*Kn = Mu/bd^2 deff. = 144 mm
Mu/bd^2 = 1.453X10^6/(1000X144^2) = 0.07 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X144/100 = 0.29 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 1.818 kN-m
Φ*Kn = Mu/bd^2 deff. = 132 mm
Mu/bd^2 = 1.818X10^6/(1000X132^2) = 0.104 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003 Reqd. Ast = 0.0003X1000X132/100 = 0.4 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

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SHEAR CHECK

(Staad Group : SHEAR)

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.067 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1684 , L/C No. - 1.0*)*

Vu = 9.65 kN

Provided Pt = 0.0045

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0045X9.648X10^3X144/1.453/10^6)X1000X144/10^3

= 146.84 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0045)X1000X144/10^3

= 147.322 kN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X144/10^3

= 247.055 kN

HENCE, Vc = 146.84 kN

Φt*Vc = 0.75X146.84 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 110.13 kN **> 9.648T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.093 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1693 , L/C No. - 1.0*)*

Vu = 12.276 kN

Pt provided = 0.0049

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0049X12.276X10^3X132/1.818/10^3)X1000X132/10^3

= 134.748 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0049)X1000X132/10^3

= 135.943 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X132/10^3

= 226.468 kN

HENCE, Vc = 134.748 kN

Φt*Vc = 0.75X134.748 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 101.061 kN **> 12.276T , Hence O.K.**

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Crack Width Check

STAAD OUTPUT FROM UNFACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	2.179	kN-m	Corresponding Mxy	-0.161	kN-m	*(Plate No. - 1681 , L/C No. - 1.0*)*
Min. Mx	-0.569	kN-m	Corresponding Mxy	-0.468	kN-m	*(Plate No. - 1679 , L/C No. - 1.0*)*
Max. My	3.794	kN-m	Corresponding Mxy	-0.161	kN-m	*(Plate No. - 1681 , L/C No. - 1.0*)*
Min. My	-1.094	kN-m	Corresponding Mxy	-0.204	kN-m	*(Plate No. - 1675 , L/C No. - 1.0*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	2.179 + 0.161 =	2.340	kN-m
My1 =	My + abs(Mxy) =	3.794 + 0.161 =	3.955	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	2.179 + 0.007 =	2.186	kN-m
My2 =	My+abs(Mxy^2/Mx) =	3.794 + 0.012 =	3.806	kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	2.34	kN-m
Myd =	3.955	kN-m

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-0.569 - 0.468 =	-1.037	kN-m
My1 =	My - abs(Mxy) =	-1.094 - 0.204 =	-1.298	kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-0.569 - 0.2 =	-0.769	kN-m
My2 =	My-abs(Mxy^2/Mx) =	-1.094 - 0.073 =	-1.021	kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd =	-1.037	kN-m
Myd =	-1.298	kN-m

CRACK WIDTH CHECK FOR BOTTOM FACE REINFORCEMENT:

E_s =	200000	Mpa	λ = 1
E_c =	27805.57498	Mpa	n = 7.1928022
f_r =	3.667969466	Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*
I_g =	0.0007	m^4	y_t = 0.1 m
M_{cr} =	24.453	kN-m	
M_{srv} =	1.298	kN-m	**O.K.**
B =	0.2152	mm^{-1}	
Neutral axis depth, c =	30.684	mm	
MOI of Cracked section, I_{cr} =	5.73E+07	mm^4	
Steel stress, f_s =	16.500	Mpa	
	16.500	Mpa	
Concrete stress, f_c =	0.695	Mpa	
s_{max} =	6323.317	mm	**O.K.** *[Max Spacing (ACI 318, eq 10-4)]*
β =	1.6712		d_c = 68 mm
A =	23800	mm^2	
w =	0.031	mm	*[ACI 224.1R-07, eq 1-1]*
w_{limit} =	0.1	mm	**O.K.**

CRACK WIDTH CHECK FOR TOP FACE REINFORCEMENT :

E_s =	200000	Mpa	λ = 1
E_c =	27805.57498	Mpa	n = 7.1928022
f_r =	3.667969466	Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*
I_g =	0.0007	m^4	y_t = 0.1 m
M_{cr} =	24.453	kN-m	
M_{srv} =	3.955	kN-m	**O.K.**
B =	0.2152	mm^{-1}	
Neutral axis depth, c =	30.684	mm	
MOI of Cracked section, I_{cr} =	5.73E+07	mm^4	
Steel stress, f_s =	50.277	Mpa	
	50.277	Mpa	
Concrete stress, f_c =	2.117	Mpa	
s_{max} =	1991.287	mm	**O.K.** *[Max Spacing (ACI 318, eq 10-4)]*
β =	1.6712		d_c = 68 mm
A =	23800	mm^2	
w =	0.093	mm	*[ACI 224.1R-07, eq 1-1]*
w_{limit} =	0.1	mm	**O.K.**

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Design of Surge Sewage Tank Wall design
(Refer staad file SURGE SEWAGE TANK.std)

For Calculation of Base pressure, Plate stress for wall, Slab Refer STAAD File

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +920.450m)
Thickness of wall	D	=	0.20	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	1144	KN/m^2	
Gross Bearing Capacity		=	1171	KN/m^2	
Earth Pressure at rest Ko		=	0.5		Ref Soil report
Vertical modulus of Subgrage Reaction in staad		=	102960	KN/m^3	Staad input for Mat Fdn.
Water Pressure for 2.2 m height			22	KN/m^2	Applied in Staad
Soil pressure for 2.0m height			18	KN/m^2	Applied in Staad
50mm Screed pressure applied on base of slab			1.2	KN/m^2	Applied in Staad
Surcharge Pressure			10	KN/m^2	Applied in Staad

1) Check for Gross Base Pressure :

Maximum Base Pressure, Pmax	=	56.0	KN/m^2	Load Case 202
Minimum Base Pressure, Pmin	=	23.0	KN/m^2	Load Case 203
				Ok

2) Design of Wall:

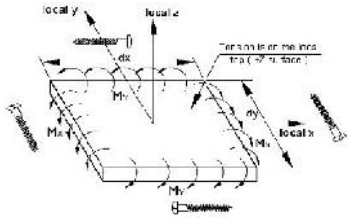
Slab Thickness =	200 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :
(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.953	kN-m	Corresponding Mxy	0.063	kN-m	*(Plate No. - 1521 , L/C No. - 1.4*)*
Min. Mx	-3.202	kN-m	Corresponding Mxy	0.287	kN-m	*(Plate No. - 1515 , L/C No. - 1.4*)*
Max. My	3.162	kN-m	Corresponding Mxy	0.063	kN-m	*(Plate No. - 1521 , L/C No. - 1.4*)*
Min. My	-4.688	kN-m	Corresponding Mxy	0.287	kN-m	*(Plate No. - 1515 , L/C No. - 1.4*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	1.953 + 0.063 =	2.016	kN-m
My1 =	My + abs(Mxy) =	3.162 + 0.063 =	3.225	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.953 + 0.001 =	1.954	kN-m
My2 =	My+abs(Mxy^2/Mx) =	3.162 + 0.002 =	3.164	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 2.016 kN-m
Myd = 3.225 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 2.016 kN-m
Φ*Kn = Mu/bd^2 deff. = 144 mm
Mu/bd^2 = 2.016X10^6/(1000X144^2) = 0.097 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003 Reqd. Ast = 0.0003X1000X144/100 = 0.43 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 3.225 kN-m
Φ*Kn = Mu/bd^2 deff. = 132 mm
Mu/bd^2 = 3.225X10^6/(1000X132^2) = 0.156 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X144/100 = 0.53 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-3.202 - 0.287 =	-3.489	kN-m
My1 =	My - abs(Mxy) =	-4.688 - 0.287 =	-4.975	kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-3.202 - 0.018 =	-3.22	kN-m
My2 =	My-abs(Mxy^2/Mx) =	-4.688 - 0.026 =	-4.662	kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -3.489 kN-m
Myd = -4.975 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 3.489 kN-m
Φ*Kn = Mu/bd^2 deff. = 144 mm
Mu/bd^2 = 3.489X10^6/(1000X144^2) = 0.168 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X144/100 = 0.58 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 4.975 kN-m
Φ*Kn = Mu/bd^2 deff. = 132 mm
Mu/bd^2 = 4.975X10^6/(1000X132^2) = 0.286 N/mm^2 < 6.9 N/mm^2
Pt = 0.0008 Reqd. Ast = 0.0008X1000X132/100 = 1.06 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

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SHEAR CHECK

(Staad Group : SHEAR)

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.08 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1513 , L/C No. - 1.0*)*

Vu = 11.52 kN

Provided Pt = 0.0045

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0045X11.52X10^3X144/3.489/10^6)X1000X144/10^3

= 141.544 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0045)X1000X144/10^3

= 147.322 kN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X144/10^3

= 247.055 kN

HENCE, Vc = 141.544 kN

Φt*Vc = 0.75X141.544 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 106.158 kN **> 11.52T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.089 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1509 , L/C No. - 1.0*)*

Vu = 11.748 kN

Pt provided = 0.0049

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0049X11.748X10^3X132/4.975/10^3)X1000X132/10^3

= 128.375 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0049)X1000X132/10^3

= 135.943 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X132/10^3

= 226.468 kN

HENCE, Vc = 128.375 kN

Φt*Vc = 0.75X128.375 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 96.281 kN **> 11.748T , Hence O.K.**

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STAAD OUTPUT FROM UNFACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.395	kN-m	Corresponding Mxy	0.045	kN-m	*(Plate No. - 1521 , L/C No. - 1.0*)*
Min. Mx	-2.287	kN-m	Corresponding Mxy	0.205	kN-m	*(Plate No. - 1515 , L/C No. - 1.0*)*
Max. My	2.258	kN-m	Corresponding Mxy	0.045	kN-m	*(Plate No. - 1521 , L/C No. - 1.0*)*
Min. My	-3.348	kN-m	Corresponding Mxy	0.205	kN-m	*(Plate No. - 1515 , L/C No. - 1.0*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	1.395 + 0.045 =	1.440 kN-m
My1 =	My + abs(Mxy) =	2.258 + 0.045 =	2.303 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.395 + 0.001 =	1.396 kN-m
My2 =	My+abs(Mxy^2/Mx) =	2.258 + 0.001 =	2.259 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	1.44 kN-m
Myd =	2.303 kN-m

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-2.287 - 0.205 =	-2.492 kN-m
My1 =	My - abs(Mxy) =	-3.348 - 0.205 =	-3.553 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-2.287 - 0.013 =	-2.3 kN-m
My2 =	My-abs(Mxy^2/Mx) =	-3.348 - 0.018 =	-3.33 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd =	-2.492 kN-m
Myd =	-3.553 kN-m

CRACK WIDTH CHECK FOR BOTTOM FACE REINFORCEMENT:

E_s =	200000 Mpa	λ = 1	
E_c =	27805.57 Mpa	n =	7.1928022
f_r =	3.67 Mpa	*[Modulus of rupture of concrete (ACI 318, eq 9-10)]*	
I_g =	0.0007 m^4	y_t =	0.1 m
M_{cr} =	24.453 kN-m		
M_{srv} =	3.553 kN-m	**O.K.**	
B =	0.2152 mm^{-1}		
Neutral axis depth, c =	30.684 mm		
MOI of Cracked section, I_{cr} =	5.73E+07 mm^4		
Steel stress, f_s =	45.166 Mpa		
	45.166 Mpa		
Concrete stress, f_c =	1.902 Mpa		
s_{max} =	2230.732 mm	**O.K.**	*[Max Spacing (ACI 318, eq 10-4)]*
β =	1.6712	d_c =	68 mm
A =	23800 mm^2		
w =	0.084 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

CRACK WIDTH CHECK FOR TOP FACE REINFORCEMENT :

E_s =	200000 Mpa	λ = 1	
E_c =	27805.6 Mpa	n =	7.1928022
f_r =	3.67 Mpa	*[Modulus of rupture of concrete (ACI 318, eq 9-10)]*	
I_g =	0.0007 m^4	y_t =	0.1 m
M_{cr} =	24.453 kN-m		
M_{srv} =	2.303 kN-m	**O.K.**	
B =	0.2152 mm^{-1}		
Neutral axis depth, c =	30.684 mm		
MOI of Cracked section, I_{cr} =	5.73E+07 mm^4		
Steel stress, f_s =	29.276 Mpa		
	29.276 Mpa		
Concrete stress, f_c =	1.233 Mpa		
s_{max} =	3509.353 mm	**O.K.**	*[Max Spacing (ACI 318, eq 10-4)]*
β =	1.6712	d_c =	68 mm
A =	23800 mm^2		
w =	0.054 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

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Design of Surge Sewage Tank Top Slab
(Refer staad file SURGE SEWAGE TANK.std)

For Calculation of Base pressure, Plate stress for wall, Slab Refer STAAD File

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +920.450m)
Thickness of Slab	D	=	0.15	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	1144	KN/m^2	
Gross Bearing Capacity		=	1171	KN/m^2	
Earth Pressure at rest Ko		=	0.5		Ref Soil report
Vertical modulus of Subgrage Reaction in staad		=	102960	KN/m^3	Staad input for Mat Fdn.
Water Pressure for 2.2 m height			22	KN/m^2	Applied in Staad
Soil pressure for 2.0m height			18	KN/m^2	Applied in Staad
50mm Screed pressure applied on base of slab			1.2	KN/m^2	Applied in Staad
Surcharge Pressure			10	KN/m^2	Applied in Staad

1) Check for Gross Base Pressure :

Maximum Base Pressure, Pmax	=	56.0	KN/m^2	Load Case 202
Minimum Base Pressure, Pmin	=	23.0	KN/m^2	Load Case 203
				Ok

2) Design of Top Slab:

Slab Thickness =	150 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	25 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

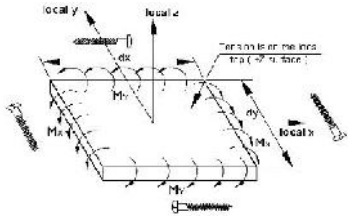
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	0.427	kN-m	Corresponding Mxy	0.335	kN-m	*(Plate No. - 1709 , L/C No. - 1.2*)*
Min. Mx	-0.998	kN-m	Corresponding Mxy	0.111	kN-m	*(Plate No. - 1711 , L/C No. - 1.2*)*
Max. My	0.883	kN-m	Corresponding Mxy	-0.003	kN-m	*(Plate No. - 1705 , L/C No. - 1.4*)*
Min. My	-1.875	kN-m	Corresponding Mxy	0.111	kN-m	*(Plate No. - 1711 , L/C No. - 1.2*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	0.427 + 0.335 =	0.762 kN-m
My1 =	My + abs(Mxy) =	0.883 + 0.003 =	0.886 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	0.427 + 0.127 =	0.554 kN-m
My2 =	My+abs(Mxy^2/Mx) =	0.883 + 0 =	0.883 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 0.762 kN-m
Myd = 0.886 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 0.762 kN-m
Φ*Kn = Mu/bd^2 deff. = 120 mm
Mu/bd^2 = 0.762X10^6/(1000X120^2) = 0.053 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X120/100 = 0.12 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0037

ALONG LOCAL Y - DIRECTION :

Mu = 0.886 kN-m
Φ*Kn = Mu/bd^2 deff. = 110 mm
Mu/bd^2 = 0.886X10^6/(1000X110^2) = 0.062 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X120/100 = 0.22 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0041

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-0.998 - 0.111 =	-1.109 kN-m
My1 =	My - abs(Mxy) =	-1.875 - 0.111 =	-1.986 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-0.998 - 0.007 =	-1.005 kN-m
My2 =	My-abs(Mxy^2/Mx) =	-1.875 - 0.012 =	-1.863 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -1.109 kN-m
Myd = -1.986 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 1.109 kN-m
Φ*Kn = Mu/bd^2 deff. = 120 mm
Mu/bd^2 = 1.109X10^6/(1000X120^2) = 0.077 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X120/100 = 0.24 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0037

ALONG LOCAL Y - DIRECTION :

Mu = 1.986 kN-m
Φ*Kn = Mu/bd^2 deff. = 110 mm
Mu/bd^2 = 1.986X10^6/(1000X110^2) = 0.164 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X110/100 = 0.44 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0041

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SHEAR CHECK

(Staad Group : SHEAR)

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.043 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1709 , L/C No. - 1.0*)*

Vu = 5.16 kN

Provided Pt = 0.0037

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0037X5.16X10^3X120/1.109/10^6)X1000X120/10^3

= 117.803 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0037)X1000X120/10^3

= 121.137 kN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X120/10^3

= 205.88 kN

HENCE, Vc = 117.803 kN

Φt*Vc = 0.75X117.803 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 88.352 kN **> 5.16T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.056 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1705 , L/C No. - 1.0*)*

Vu = 6.16 kN

Pt provided = 0.0041

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0041X6.16X10^3X110/1.986/10^3)X1000X110/10^3

= 106.739 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0041)X1000X110/10^3

= 111.79 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X110/10^3

= 188.723 kN

HENCE, Vc = 106.739 kN

Φt*Vc = 0.75X106.739 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 80.054 kN **> 6.16T , Hence O.K.**

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Crack Width Check

STAAD OUTPUT FROM UNFACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	0.288	kN-m	Corresponding Mxy	0.254	kN-m	*(Plate No. - 1709 , L/C No. - 1.0*)*
Min. Mx	-0.790	kN-m	Corresponding Mxy	0.082	kN-m	*(Plate No. - 1711 , L/C No. - 1.0*)*
Max. My	0.631	kN-m	Corresponding Mxy	-0.002	kN-m	*(Plate No. - 1705 , L/C No. - 1.0*)*
Min. My	-1.429	kN-m	Corresponding Mxy	0.082	kN-m	*(Plate No. - 1711 , L/C No. - 1.0*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	0.288 + 0.254 =	0.542 kN-m
My1 =	My + abs(Mxy) =	0.631 + 0.002 =	0.633 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	0.288 + 0.102 =	0.39 kN-m
My2 =	My+abs(Mxy^2/Mx) =	0.631 + 0 =	0.631 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	0.542 kN-m
Myd =	0.633 kN-m

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-0.79 - 0.082 =	-0.872 kN-m
My1 =	My - abs(Mxy) =	-1.429 - 0.082 =	-1.511 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-0.79 - 0.005 =	-0.795 kN-m
My2 =	My-abs(Mxy^2/Mx) =	-1.429 - 0.009 =	-1.42 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd =	-0.872 kN-m
Myd =	-1.511 kN-m

CRACK WIDTH CHECK FOR BOTTOM FACE REINFORCEMENT:

E_s =	200000 Mpa	λ = 1	
E_c =	27805.57498 Mpa	n = 7.1928022	
f_r =	3.667969466 Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*	
I_g =	0.0003 m^4	y_t = 0.075 m	
M_{cr} =	13.755 kN-m		
M_{srv} =	1.511 kN-m	**O.K.**	
B =	0.3096 mm^{-1}		
Neutral axis depth, c =	23.621 mm		
MOI of Cracked section, I_{cr} =	2.85E+07 mm^4		
Steel stress, f_s =	32.952 Mpa		
	32.952 Mpa		
Concrete stress, f_c =	1.253 Mpa		
s_{max} =	3166.453 mm	**O.K.**	*[Max Spacing (ACI 318, eq 10-4)]*
β =	1.4631	d_c = 40 mm	
A =	14000 mm^2		
w =	0.046 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

CRACK WIDTH CHECK FOR TOP FACE REINFORCEMENT :

E_s =	200000 Mpa	λ = 1	
E_c =	27805.57498 Mpa	n = 7.1928022	
f_r =	3.667969466 Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*	
I_g =	0.0003 m^4	y_t = 0.075 m	
M_{cr} =	13.755 kN-m		
M_{srv} =	0.633 kN-m	**O.K.**	
B =	0.3096 mm^{-1}		
Neutral axis depth, c =	23.621 mm		
MOI of Cracked section, I_{cr} =	2.85E+07 mm^4		
Steel stress, f_s =	13.804 Mpa		
	13.804 Mpa		
Concrete stress, f_c =	0.525 Mpa		
s_{max} =	7645.159 mm	**O.K.**	*[Max Spacing (ACI 318, eq 10-4)]*
β =	1.4631	d_c = 40 mm	
A =	14000 mm^2		
w =	0.019 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

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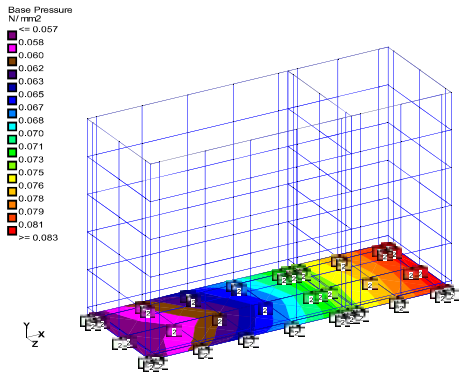
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8. DESIGN CALCULATIONS FOR SUMP TANK

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Design of Sump Tank Base Slab
(Refer staad file SUMP TANK.std)



For Calculation of Base pressure, Plate stress for wall, Slab Refer STAAD File

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +920.450m)
Thickness of Base Slab	D	=	0.20	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	1144	KN/m^2	
Gross Bearing Capacity		=	1171	KN/m^2	
Earth Pressure at rest Ko		=	0.5		Ref Soil report
Vertical modulus of Subgrage Reaction in staad		=	102960	KN/m^3	Staad input for Mat Fdn.
Water Pressure for 2.5 m height			25	KN/m^2	Applied in Staad
Soil pressure for 2.0m height			18	KN/m^2	Applied in Staad
50mm Screed pressure applied on base of slab			1.2	KN/m^2	Applied in Staad
Surcharge Pressure			10	KN/m^2	Applied in Staad

1) Check for Gross Base Pressure :

Maximum Base Pressure, Pmax	=	83.0	KN/m^2	Load Case 206
Minimum Base Pressure, Pmin	=	32.0	KN/m^2	Load Case 205
				Ok

2) Design of Base Slab :

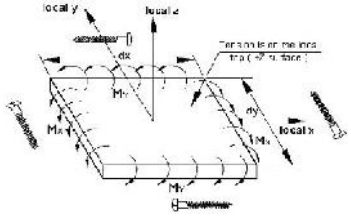
Slab Thickness =	200 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.560	kN-m	Corresponding Mxy	0.341	kN-m	*(Plate No. - 1675 , L/C No. - 1.2*)*
Min. Mx	-0.010	kN-m	Corresponding Mxy	0.647	kN-m	*(Plate No. - 1676 , L/C No. - 1.4*)*
Max. My	3.287	kN-m	Corresponding Mxy	0.341	kN-m	*(Plate No. - 1675 , L/C No. - 1.2*)*
Min. My	0.636	kN-m	Corresponding Mxy	0.221	kN-m	*(Plate No. - 1678 , L/C No. - 1.2*)*

FOR TOP FACE REINFORCEMRNT :



Mx1 =	Mx + abs(Mxy) =	1.56 + 0.341 =	1.901	kN-m
My1 =	My + abs(Mxy) =	3.287 + 0.341 =	3.628	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.56 + 0.035 =	1.595	kN-m
My2 =	My+abs(Mxy^2/Mx) =	3.287 + 0.075 =	3.362	kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 1.901 kN-m
Myd = 3.628 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 1.901 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 1.901X10^6/(1000X144^2) = 0.092 N/mm^2 < deff. = 144 mm 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X144/100 = 0.29 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 3.628 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 3.628X10^6/(1000X132^2) = 0.175 N/mm^2 < deff. = 132 mm 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X144/100 = 0.66 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-0.01 - 0.647 =	-0.657	kN-m
My1 =	My - abs(Mxy) =	0.636 - 0.221 =	0.415	kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-0.01 - 0.658 =	-0.668	kN-m
My2 =	My-abs(Mxy^2/Mx) =	0.636 - 4.884 =	5.52	kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -0.668 kN-m
Myd = 0.001 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 0.668 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 0.668X10^6/(1000X144^2) = 0.032 N/mm^2 < deff. = 144 mm 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X144/100 = 0.14 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 0.001 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 0.001X10^6/(1000X132^2) = 0 N/mm^2 < deff. = 132 mm 6.9 N/mm^2
Pt = 0.0000 Reqd. Ast = 0X1000X132/100 = 0 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

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SHEAR CHECK

(Staad Group : SHEAR)

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.052 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1676 , L/C No. - 1.2*)*

Vu = 7.49 kN

Provided Pt = 0.0045

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0045X7.488X10^3X144/0.668/10^6)X1000X144/10^3

= 154.088 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0045)X1000X144/10^3

= 147.322 kN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X144/10^3

= 247.055 kN

HENCE, Vc = 147.322 kN

Φt*Vc = 0.75X147.322 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 110.492 kN **> 7.488T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.06 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1693 , L/C No. - 1.2*)*

Vu = 7.92 kN

Pt provided = 0.0049

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0049X7.92X10^3X132/0.001/10^3)X1000X132/10^3

= 11620.188 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0049)X1000X132/10^3

= 135.943 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X132/10^3

= 226.468 kN

HENCE, Vc = 135.943 kN

Φt*Vc = 0.75X135.943 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 101.957 kN **> 7.92T , Hence O.K.**

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Crack Width Check

STAAD OUTPUT FROM UNFACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.233	kN-m	Corresponding Mxy	0.280	kN-m	*(Plate No. - 1675 , L/C No. - 1.0*)*
Min. Mx	-0.007	kN-m	Corresponding Mxy	0.462	kN-m	*(Plate No. - 1676 , L/C No. - 1.0*)*
Max. My	2.630	kN-m	Corresponding Mxy	0.280	kN-m	*(Plate No. - 1675 , L/C No. - 1.0*)*
Min. My	0.454	kN-m	Corresponding Mxy	0.158	kN-m	*(Plate No. - 1678 , L/C No. - 1.0*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	1.233 + 0.28 =	1.513 kN-m
My1 =	My + abs(Mxy) =	2.63 + 0.28 =	2.910 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.233 + 0.03 =	1.263 kN-m
My2 =	My+abs(Mxy^2/Mx) =	2.63 + 0.064 =	2.694 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	1.513 kN-m
Myd =	2.91 kN-m

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-0.007 - 0.462 =	-0.469 kN-m
My1 =	My - abs(Mxy) =	0.454 - 0.158 =	0.296 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-0.007 - 0.47 =	-0.477 kN-m
My2 =	My-abs(Mxy^2/Mx) =	0.454 - 3.566 =	4.02 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd =	-0.477 kN-m
Myd =	0 kN-m

CRACK WIDTH CHECK FOR BOTTOM FACE REINFORCEMENT:

E_s =	200000 Mpa	λ = 1	
E_c =	27805.57498 Mpa	n = 7.1928022	
f_r =	3.667969466 Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*	
I_g =	0.0007 m^4	y_t = 0.1 m	
M_{cr} =	24.453 kN-m		
M_{srv} =	0 kN-m	**O.K.**	
B =	0.2152 mm^{-1}		
Neutral axis depth, c =	30.684 mm		
MOI of Cracked section, I_{cr} =	5.73E+07 mm^4		
Steel stress, f_s =	0.010 Mpa		
	0.000 Mpa		
Concrete stress, f_c =	0.000 Mpa		
s_{max} =	10639875 mm	**O.K.**	*[Max Spacing (ACI 318, eq 10-4)]*
β =	1.6712	d_c = 68 mm	
A =	23800 mm^2		
w =	0.000 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

CRACK WIDTH CHECK FOR TOP FACE REINFORCEMENT :

E_s =	200000 Mpa	λ = 1	
E_c =	27805.57498 Mpa	n = 7.1928022	
f_r =	3.667969466 Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*	
I_g =	0.0007 m^4	y_t = 0.1 m	
M_{cr} =	24.453 kN-m		
M_{srv} =	2.91 kN-m	**O.K.**	
B =	0.2152 mm^{-1}		
Neutral axis depth, c =	30.684 mm		
MOI of Cracked section, I_{cr} =	5.73E+07 mm^4		
Steel stress, f_s =	36.992 Mpa		
	36.992 Mpa		
Concrete stress, f_c =	1.558 Mpa		
s_{max} =	2751.259 mm	**O.K.**	*[Max Spacing (ACI 318, eq 10-4)]*
β =	1.6712	d_c = 68 mm	
A =	23800 mm^2		
w =	0.069 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

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Design of Sump Tank Wall design
(Refer staad file SUMP TANK.std)

For Calculation of Base pressure, Plate stress for wall, Slab Refer STAAD File

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +920.450m)
Thickness of wall	D	=	0.20	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	1144	KN/m^2	
Gross Bearing Capacity		=	1171	KN/m^2	
Earth Pressure at rest Ko		=	0.5		Ref Soil report
Vertical modulus of Subgrage Reaction in staad		=	102960	KN/m^3	Staad input for Mat Fdn.
Water Pressure for 2.5 m height			25	KN/m^2	Applied in Staad
Soil pressure for 2.0m height			18	KN/m^2	Applied in Staad
50mm Screed pressure applied on base of slab			1.2	KN/m^2	Applied in Staad
Surcharge Pressure			10	KN/m^2	Applied in Staad

1) Check for Gross Base Pressure :

Maximum Base Pressure, Pmax	=	83.0	KN/m^2	Load Case 206
Minimum Base Pressure, Pmin	=	32.0	KN/m^2	Load Case 205
				Ok

2) Design of Wall:

Slab Thickness =	200 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

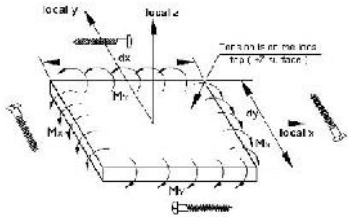
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.669	kN-m	Corresponding Mxy	-0.006	kN-m	*(Plate No. - 1669 , L/C No. - 1.4*)*
Min. Mx	-3.221	kN-m	Corresponding Mxy	-0.079	kN-m	*(Plate No. - 1520 , L/C No. - 1.4*)*
Max. My	0.460	kN-m	Corresponding Mxy	-0.128	kN-m	*(Plate No. - 1665 , L/C No. - 1.4*)*
Min. My	-2.328	kN-m	Corresponding Mxy	0.331	kN-m	*(Plate No. - 1514 , L/C No. - 1.4*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	1.669 + 0.006 =	1.675	kN-m
My1 =	My + abs(Mxy) =	0.46 + 0.128 =	0.588	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.669 + 0 =	1.669	kN-m
My2 =	My+abs(Mxy^2/Mx) =	0.46 + 0.01 =	0.47	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 1.675 kN-m
Myd = 0.588 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 1.675 kN-m
Φ*Kn = Mu/bd^2 deff. = 144 mm
Mu/bd^2 = 1.675X10^6/(1000X144^2) = 0.081 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X144/100 = 0.29 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 0.588 kN-m
Φ*Kn = Mu/bd^2 deff. = 132 mm
Mu/bd^2 = 0.588X10^6/(1000X132^2) = 0.028 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X144/100 = 0.13 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

FOR BOTTOM FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-3.221 - 0.079 =	-3.3	kN-m
My1 =	My - abs(Mxy) =	-2.328 - 0.331 =	-2.659	kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-3.221 - 0.003 =	-3.224	kN-m
My2 =	My-abs(Mxy^2/Mx) =	-2.328 - 0.034 =	-2.294	kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -3.3 kN-m
Myd = -2.659 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 3.3 kN-m
Φ*Kn = Mu/bd^2 deff. = 144 mm
Mu/bd^2 = 3.3X10^6/(1000X144^2) = 0.159 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X144/100 = 0.58 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0045

ALONG LOCAL Y - DIRECTION :

Mu = 2.659 kN-m
Φ*Kn = Mu/bd^2 deff. = 132 mm
Mu/bd^2 = 2.659X10^6/(1000X132^2) = 0.153 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X132/100 = 0.53 cm^2
Provided **12** mm @ **175** C/C (Provide Ast = 6.46)cm^2

Total Provide Ast = 6.46 cm^2 **O.K.**
Total Provide Pt = 0.0049

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SHEAR CHECK

(Staad Group : SHEAR)

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.069 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1591 , L/C No. - 1.4*)*

Vu = 9.94 kN

Provided Pt = 0.0045

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0045X9.936X10^3X144/3.3/10^6)X1000X144/10^3

= 141.083 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0045)X1000X144/10^3

= 147.322 kN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X144/10^3

= 247.055 kN

HENCE, Vc = 141.083 kN

Φt*Vc = 0.75X141.083 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 105.812 kN **> 9.936T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.045 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1508 , L/C No. - 1.4*)*

Vu = 5.94 kN

Pt provided = 0.0049

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0049X5.94X10^3X132/2.659/10^3)X1000X132/10^3

= 128.19 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0049)X1000X132/10^3

= 135.943 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X132/10^3

= 226.468 kN

HENCE, Vc = 128.19 kN

Φt*Vc = 0.75X128.19 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 96.143 kN **> 5.94T , Hence O.K.**

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STAAD OUTPUT FROM UNFACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.219	kN-m	Corresponding Mxy	-0.001	kN-m	*(Plate No. - 1669 , L/C No. - 1.0*)*
Min. Mx	-2.334	kN-m	Corresponding Mxy	-0.065	kN-m	*(Plate No. - 1520 , L/C No. - 1.0*)*
Max. My	0.338	kN-m	Corresponding Mxy	-0.090	kN-m	*(Plate No. - 1665 , L/C No. - 1.0*)*
Min. My	-1.710	kN-m	Corresponding Mxy	0.235	kN-m	*(Plate No. - 1514 , L/C No. - 1.0*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	1.219 + 0.001 =	1.220 kN-m
My1 =	My + abs(Mxy) =	0.338 + 0.09 =	0.428 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.219 + 0 =	1.219 kN-m
My2 =	My+abs(Mxy^2/Mx) =	0.338 + 0.007 =	0.345 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 1.22 kN-m
Myd = 0.428 kN-m

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	-2.334 - 0.065 =	-2.399 kN-m
My1 =	My - abs(Mxy) =	-1.71 - 0.235 =	-1.945 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-2.334 - 0.002 =	-2.336 kN-m
My2 =	My-abs(Mxy^2/Mx) =	-1.71 - 0.024 =	-1.686 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -2.399 kN-m
Myd = -1.945 kN-m

CRACK WIDTH CHECK FOR BOTTOM FACE REINFORCEMENT:

E_s = 200000 Mpa — λ = 1
E_c = 27805.57 Mpa — n = 7.1928022
f_r = 3.67 Mpa — *[Modulus of rupture of concrete (ACI 318, eq 9-10)]*
I_g = 0.0007 m^4 — y_t = 0.1 m
M_{cr} = 24.453 kN-m
M_{srv} = 1.945 kN-m — **O.K.**

B = 0.2152 mm^{-1}
Neutral axis depth, c = 30.684 mm
MOI of Cracked section, I_{cr} = 5.73E+07 mm^4
Steel stress, f_s = 24.725 Mpa
24.725 Mpa
Concrete stress, f_c = 1.041 Mpa

s_{max} = 4178.298 mm — **O.K.** *[Max Spacing (ACI 318, eq 10-4)]*

β = 1.6712 — d_c = 68 mm
A = 23800 mm^2
w = 0.046 mm — *[ACI 224.1R-07, eq 1-1]*
w_{limit} = 0.1 mm — **O.K.**

CRACK WIDTH CHECK FOR TOP FACE REINFORCEMENT :

E_s = 200000 Mpa — λ = 1
E_c = 27805.6 Mpa — n = 7.1928022
f_r = 3.67 Mpa — *[Modulus of rupture of concrete (ACI 318, eq 9-10)]*
I_g = 0.0007 m^4 — y_t = 0.1 m
M_{cr} = 24.453 kN-m
M_{srv} = 0.428 kN-m — **O.K.**

B = 0.2152 mm^{-1}
Neutral axis depth, c = 30.684 mm
MOI of Cracked section, I_{cr} = 5.73E+07 mm^4
Steel stress, f_s = 5.441 Mpa
5.441 Mpa
Concrete stress, f_c = 0.229 Mpa

s_{max} = 19430.876 mm — **O.K.** *[Max Spacing (ACI 318, eq 10-4)]*

β = 1.6712 — d_c = 68 mm
A = 23800 mm^2
w = 0.010 mm — *[ACI 224.1R-07, eq 1-1]*
w_{limit} = 0.1 mm — **O.K.**

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Design of Sump Tank (Waffle Wall)
(Refer staad file SUMP TANK.std)

For Calculation of Base pressure, Plate stress for wall, Slab Refer STAAD File

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +920.450m)
Thickness of Wall	D	=	0.20	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	1144	KN/m^2	
Gross Bearing Capacity		=	1171	KN/m^2	
Earth Pressure at rest Ko		=	0.5		Ref Soil report
Vertical modulus of Subgrage Reaction in staad		=	102960	KN/m^3	Staad input for Mat Fdn.
Water Pressure for 2.5 m height			25	KN/m^2	Applied in Staad
Soil pressure for 2.0m height			18	KN/m^2	Applied in Staad
50mm Screed pressure applied on base of slab			1.2	KN/m^2	Applied in Staad
Surcharge Pressure			10	KN/m^2	Applied in Staad

1) Check for Gross Base Pressure :

Maximum Base Pressure, Pmax	=	83.0	KN/m^2	Load Case 206
Minimum Base Pressure, Pmin	=	32.0	KN/m^2	Load Case 205
				Ok

2) Design of Top Slab:

Slab Thickness =	200 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	25 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

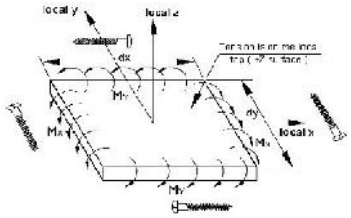
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.436	kN-m	Corresponding Mxy	0.032	kN-m	*(Plate No. - 1783 , L/C No. - 1.0*)*
Min. Mx	0.147	kN-m	Corresponding Mxy	-0.083	kN-m	*(Plate No. - 1775 , L/C No. - 1.0*)*
Max. My	0.592	kN-m	Corresponding Mxy	-0.081	kN-m	*(Plate No. - 1779 , L/C No. - 1.0*)*
Min. My	-0.224	kN-m	Corresponding Mxy	-0.083	kN-m	*(Plate No. - 1775 , L/C No. - 1.0*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	1.436 + 0.032 =	1.468 kN-m
My1 =	My + abs(Mxy) =	0.592 + 0.081 =	0.673 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.436 + 0.002 =	1.438 kN-m
My2 =	My+abs(Mxy^2/Mx) =	0.592 + 0.007 =	0.599 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 1.468 kN-m
Myd = 0.673 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 1.468 kN-m
Φ*Kn = Mu/bd^2 deff. = 170 mm
Mu/bd^2 = 1.468X10^6/(1000X170^2) = 0.051 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X170/100 = 0.17 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0026

ALONG LOCAL Y - DIRECTION :

Mu = 0.673 kN-m
Φ*Kn = Mu/bd^2 deff. = 160 mm
Mu/bd^2 = 0.673X10^6/(1000X160^2) = 0.023 N/mm^2 < 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X170/100 = 0.16 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0028

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	0.147 - 0.083 =	0.064 kN-m
My1 =	My - abs(Mxy) =	-0.224 - 0.083 =	-0.307 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	0.147 - 0.031 =	0.116 kN-m
My2 =	My-abs(Mxy^2/Mx) =	-0.224 - 0.047 =	-0.177 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = 0.001 kN-m
Myd = -0.177 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 0.001 kN-m
Φ*Kn = Mu/bd^2 deff. = 170 mm
Mu/bd^2 = 0.001X10^6/(1000X170^2) = 0 N/mm^2 < 6.9 N/mm^2
Pt = 0.0000 Reqd. Ast = 0X1000X170/100 = 0 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0026

ALONG LOCAL Y - DIRECTION :

Mu = 0.177 kN-m
Φ*Kn = Mu/bd^2 deff. = 160 mm
Mu/bd^2 = 0.177X10^6/(1000X160^2) = 0.007 N/mm^2 < 6.9 N/mm^2
Pt = 0.0000 Reqd. Ast = 0X1000X160/100 = 0 cm^2
Provided **10** mm @ **175** C/C (Provide Ast = 4.49)cm^2

Total Provide Ast = 4.49 cm^2 **O.K.**
Total Provide Pt = 0.0028

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SHEAR CHECK

(Staad Group : SHEAR)

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.057 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1782 , L/C No. - 1.0*)*

Vu = 9.69 kN

Provided Pt = 0.0026

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X9.69X10^3X170/0.001/10^6)X1000X170/10^3

= 12538.73 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X170/10^3

= 168.431 kN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X170/10^3

= 291.663 kN

HENCE, Vc = 168.431 kN

Φt*Vc = 0.75X168.431 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 126.323 kN **> 9.69T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.033 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1775 , L/C No. - 1.0*)*

Vu = 5.28 kN

Pt provided = 0.0028

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[ACI 318-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0028X5.28X10^3X160/0.177/10^3)X1000X160/10^3

= 187.802 kN

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0028)X1000X160/10^3

= 159.068 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X160/10^3

= 274.506 kN

HENCE, Vc = 159.068 kN

Φt*Vc = 0.75X159.068 *[ACI 318-CR-18, CL. No. 7.6.3.1, Page-42]*

= 119.301 kN **> 5.28T , Hence O.K.**

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Crack Width Check

STAAD OUTPUT FROM UNFACTORED LOAD COMBINATION :

(Staad Group : MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.111	kN-m	Corresponding Mxy	0.014	kN-m	*(Plate No. - 1783 , L/C No. - 1.0*)*
Min. Mx	0.105	kN-m	Corresponding Mxy	-0.059	kN-m	*(Plate No. - 1775 , L/C No. - 1.0*)*
Max. My	0.427	kN-m	Corresponding Mxy	-0.059	kN-m	*(Plate No. - 1779 , L/C No. - 1.0*)*
Min. My	-0.160	kN-m	Corresponding Mxy	-0.059	kN-m	*(Plate No. - 1775 , L/C No. - 1.0*)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	1.111 + 0.014 =	1.125 kN-m
My1 =	My + abs(Mxy) =	0.427 + 0.059 =	0.486 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	1.111 + 0 =	1.111 kN-m
My2 =	My+abs(Mxy^2/Mx) =	0.427 + 0.005 =	0.432 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	1.125 kN-m
Myd =	0.486 kN-m

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 =	Mx - abs(Mxy) =	0.105 - 0.059 =	0.046 kN-m
My1 =	My - abs(Mxy) =	-0.16 - 0.059 =	-0.219 kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	0.105 - 0.022 =	0.083 kN-m
My2 =	My-abs(Mxy^2/Mx) =	-0.16 - 0.033 =	-0.127 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd =	0 kN-m
Myd =	-0.127 kN-m

CRACK WIDTH CHECK FOR BOTTOM FACE REINFORCEMENT:

E_s =	200000 Mpa	λ = 1	
E_c =	27805.57498 Mpa	n = 7.1928022	
f_r =	3.667969466 Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*	
I_g =	0.0007 m^4	y_t = 0.1 m	
M_{cr} =	24.453 kN-m		
M_{srv} =	0.127 kN-m	**O.K.**	
B =	0.3096 mm^{-1}		
Neutral axis depth, c =	29.080 mm		
MOI of Cracked section, I_{cr} =	6.36E+07 mm^4		
Steel stress, f_s =	1.882 Mpa		
	1.882 Mpa		
Concrete stress, f_c =	0.058 Mpa		
s_{max} =	56478.402 mm	**O.K.** *[Max Spacing (ACI 318, eq 10-4)]*	
β =	1.3055	d_c = 40 mm	
A =	14000 mm^2		
w =	0.002 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

CRACK WIDTH CHECK FOR TOP FACE REINFORCEMENT :

E_s =	200000 Mpa	λ = 1	
E_c =	27805.57498 Mpa	n = 7.1928022	
f_r =	3.667969466 Mpa	*[Modulus of rupture of cocrete (ACI 318, eq 9-10)]*	
I_g =	0.0007 m^4	y_t = 0.1 m	
M_{cr} =	24.453 kN-m		
M_{srv} =	0.486 kN-m	**O.K.**	
B =	0.3096 mm^{-1}		
Neutral axis depth, c =	29.080 mm		
MOI of Cracked section, I_{cr} =	6.36E+07 mm^4		
Steel stress, f_s =	7.201 Mpa		
	7.201 Mpa		
Concrete stress, f_c =	0.222 Mpa		
s_{max} =	14712.592 mm	**O.K.** *[Max Spacing (ACI 318, eq 10-4)]*	
β =	1.3055	d_c = 40 mm	
A =	14000 mm^2		
w =	0.009 mm	*[ACI 224.1R-07, eq 1-1]*	
w_{limit} =	0.1 mm	**O.K.**	

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectual Design Criteria (Doc. MD-216-1000-EG-ST-CRT-1001) Clause 4.1

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ATTACHMENT 1 STAAD INPUT FOR SURGE SEWAGE TANK

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 11-Sep-20
ENGINEER NAME BJB
CHECKER NAME SJM
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 3.2 0 0; 3 0 0 2.2; 4 3.2 0 2.2; 5 0 -2.1 0; 6 3.2 -2.1 0;
7 0 -2.1 2.2; 8 3.2 -2.1 2.2; 9 0 0.2 0; 10 3.2 0.2 0; 11 0 0.2 2.2;
12 3.2 0.2 2.2; 13 0 -2 0; 14 3.2 -2 0; 15 0 -2 2.2; 16 3.2 -2 2.2; 17 0.1 0 0;
18 0.1 0 2.2; 19 0.1 -2.1 0; 20 0.1 -2.1 2.2; 21 0.1 0.2 0; 22 0.1 0.2 2.2;
23 0.1 -2 0; 24 0.1 -2 2.2; 25 3.1 0 0; 26 3.1 0 2.2; 27 3.1 -2.1 0;
28 3.1 -2.1 2.2; 29 3.1 0.2 0; 30 3.1 0.2 2.2; 31 3.1 -2 0; 32 3.1 -2 2.2;
33 0 0 0.1; 34 3.2 0 0.1; 35 0 -2.1 0.1; 36 3.2 -2.1 0.1; 37 0 0.2 0.1;
38 3.2 0.2 0.1; 39 0 -2 0.1; 40 3.2 -2 0.1; 41 0 0 2.1; 42 3.2 0 2.1;
43 0 -2.1 2.1; 44 3.2 -2.1 2.1; 45 0 0.2 2.1; 46 3.2 0.2 2.1; 47 0 -2 2.1;
48 3.2 -2 2.1; 50 0.1 -2.1 0.1; 54 3.1 -2.1 0.1; 58 0.1 -2.1 2.1;
62 3.1 -2.1 2.1; 943 0 0.275 0; 944 3.2 0.275 0; 945 0 0.275 2.2;
946 3.2 0.275 2.2; 947 0.1 0.275 0; 948 0.1 0.275 2.2; 949 3.1 0.275 0;
950 3.1 0.275 2.2; 951 0 0.275 0.1; 952 3.2 0.275 0.1; 953 0 0.275 2.1;
954 3.2 0.275 2.1; 1013 0.1 0.275 0.1; 1014 3.1 0.275 0.1; 1015 0.1 0.275 2.1;
1016 3.1 0.275 2.1; 1029 3.1 -2.1 0.6; 1030 3.2 -2.1 0.6; 1031 3.1 -2.1 1.1;
1032 3.2 -2.1 1.1; 1033 3.1 -2.1 1.6; 1034 3.2 -2.1 1.6; 1035 3.2 -2 0.6;
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1724 1252 1253 1238 1237; 1725 1253 1254 1239 1238; 1726 1254 1181 1014 1239;
START GROUP DEFINITION
ELEMENT
_WALL 1507 TO 1536 1551 TO 1570 1583 TO 1612 1637 TO 1640 1657 TO 1672
_TOP_SLAB 1703 TO 1726
_BOT_SLAB 1673 TO 1696
END GROUP DEFINITION
ELEMENT PROPERTY
61 TO 76 1181 TO 1188 1471 TO 1474 1479 TO 1482 1489 TO 1494 1501 TO 1618 -
1633 TO 1672 THICKNESS 0.2
977 TO 980 1467 TO 1470 1475 TO 1478 1483 TO 1488 1495 TO 1500 1673 TO 1695 -
1696 THICKNESS 0.2
1240 TO 1243 1619 TO 1632 1697 TO 1726 THICKNESS 0.15
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 35000
END DEFINE MATERIAL
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
977 TO 980 1467 TO 1470 1475 TO 1478 1483 TO 1488 1495 TO 1500 1673 TO 1695 -
1696 PLATE MAT DIRECT Y SUBGRADE 102960 PRINT
LOAD 1 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
1673 TO 1696 PR -1.2
LOAD 2 LOADTYPE Live  TITLE LIVE LOAD
ELEMENT LOAD
1703 TO 1726 PR -5
LOAD 3 LOADTYPE Fluids  TITLE WATER PRESSURE
ELEMENT LOAD
1507 TO 1512 1551 TO 1554 1583 TO 1588 1657 TO 1660 TRAP JT -22 -22 -17 -17
1513 TO 1518 1555 TO 1558 1589 TO 1594 1661 TO 1664 TRAP JT -17 -17 -12 -12
1519 TO 1524 1559 TO 1562 1595 TO 1600 1665 TO 1668 TRAP JT -12 -12 -7 -7
1525 TO 1530 1563 TO 1566 1601 TO 1606 1669 TO 1672 TRAP JT -7 -7 -2 -2
1531 TO 1536 1567 TO 1570 1607 TO 1612 1637 TO 1639 -
1640 TRAP JT -2 -2 -4.47035e-007 -4.47035e-007
1673 TO 1696 PR -22
LOAD 4 LOADTYPE Soil  TITLE SOIL PRESSURE
ELEMENT LOAD
1507 TO 1512 1543 1547 1551 TO 1554 1575 1579 1583 TO 1588 1641 1645 1649 -
1653 1657 TO 1660 TRAP JT 18 18 13.5 13.5
1513 TO 1518 1544 1548 1555 TO 1558 1576 1580 1589 TO 1594 1642 1646 1650 -
1654 1661 TO 1664 TRAP JT 13.5 13.5 9 9
1519 TO 1524 1545 1549 1559 TO 1562 1577 1581 1595 TO 1600 1643 1647 1651 -
1655 1665 TO 1668 TRAP JT 9 9 4.5 4.5
1525 TO 1530 1546 1550 1563 TO 1566 1578 1582 1601 TO 1606 1644 1648 1652 -
1656 1669 TO 1672 TRAP JT 4.5 4.5 0 0
1507 TO 1530 1543 TO 1566 1575 TO 1606 1641 TO 1672 PR 5
******FACTORED LOAD COMBINATIONS*****
LOAD COMB 101 1.4*(D+F)
1 1.4 3 1.4
LOAD COMB 102 1.2*(D+F)+1.6*(L+H)
1 1.2 3 1.2 2 1.6 4 1.6
LOAD COMB 103 1.4*(D+H)
1 1.4 4 1.4
******UNFACTORED LOAD COMBINATIONS*****
LOAD COMB 201 1.0*(D+F)
1 1.0 3 1.0
LOAD COMB 202 1.0*(D+F)+1.0*(L+H)
1 1.0 3 1.0 2 1.0 4 1.0
LOAD COMB 203 1.0*(D+H)
1 1.0 4 1.0
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 2 STAAD OUTPUT FOR SURGE SEWAGE TANK

PROPRIETARY & CONFIDENTIAL

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    OCT 15, 2020                *
*           Time=     10:35: 8                   *
*                                                *
*      USER ID: LNT ECC DIV                      *
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    1. STAAD SPACE
INPUT FILE: D:\MAADEN\Sewage Treatment Plant\Surge Sewage Tank\STAAD\SURGE SEWAGE TANK.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 11-SEP-20
    4. ENGINEER NAME BJB
    5. CHECKER NAME SJM
    6. END JOB INFORMATION
    7. INPUT WIDTH 79
    8. UNIT METER KN
    9. JOINT COORDINATES
   10. 1 0 0 0; 2 3.2 0 0; 3 0 0 2.2; 4 3.2 0 2.2; 5 0 -2.1 0; 6 3.2 -2.1 0
   11. 7 0 -2.1 2.2; 8 3.2 -2.1 2.2; 9 0 0.2 0; 10 3.2 0.2 0; 11 0 0.2 2.2
   12. 12 3.2 0.2 2.2; 13 0 -2 0; 14 3.2 -2 0; 15 0 -2 2.2; 16 3.2 -2 2.2; 17 0.1 0 0
   13. 18 0.1 0 2.2; 19 0.1 -2.1 0; 20 0.1 -2.1 2.2; 21 0.1 0.2 0; 22 0.1 0.2 2.2
   14. 23 0.1 -2 0; 24 0.1 -2 2.2; 25 3.1 0 0; 26 3.1 0 2.2; 27 3.1 -2.1 0
   15. 28 3.1 -2.1 2.2; 29 3.1 0.2 0; 30 3.1 0.2 2.2; 31 3.1 -2 0; 32 3.1 -2 2.2
   16. 33 0 0 0.1; 34 3.2 0 0.1; 35 0 -2.1 0.1; 36 3.2 -2.1 0.1; 37 0 0.2 0.1
   17. 38 3.2 0.2 0.1; 39 0 -2 0.1; 40 3.2 -2 0.1; 41 0 0 2.1; 42 3.2 0 2.1
   18. 43 0 -2.1 2.1; 44 3.2 -2.1 2.1; 45 0 0.2 2.1; 46 3.2 0.2 2.1; 47 0 -2 2.1
   19. 48 3.2 -2 2.1; 50 0.1 -2.1 0.1; 54 3.1 -2.1 0.1; 58 0.1 -2.1 2.1
   20. 62 3.1 -2.1 2.1; 943 0 0.275 0; 944 3.2 0.275 0; 945 0 0.275 2.2
   21. 946 3.2 0.275 2.2; 947 0.1 0.275 0; 948 0.1 0.275 2.2; 949 3.1 0.275 0
   22. 950 3.1 0.275 2.2; 951 0 0.275 0.1; 952 3.2 0.275 0.1; 953 0 0.275 2.1
   23. 954 3.2 0.275 2.1; 1013 0.1 0.275 0.1; 1014 3.1 0.275 0.1; 1015 0.1 0.275 2.1
   24. 1016 3.1 0.275 2.1; 1029 3.1 -2.1 0.6; 1030 3.2 -2.1 0.6; 1031 3.1 -2.1 1.1
   25. 1032 3.2 -2.1 1.1; 1033 3.1 -2.1 1.6; 1034 3.2 -2.1 1.6; 1035 3.2 -2 0.6
   26. 1036 3.2 -2 1.1; 1037 3.2 -2 1.6; 1038 0.1 -2.1 1.6; 1039 0 -2.1 1.6
   27. 1040 0.1 -2.1 1.1; 1041 0 -2.1 1.1; 1042 0.1 -2.1 0.6; 1043 0 -2.1 0.6
   28. 1044 0 -2 1.6; 1045 0 -2 1.1; 1046 0 -2 0.6; 1047 0.6 -2.1 2.2
   29. 1048 0.6 -2.1 2.1; 1049 1.1 -2.1 2.2; 1050 1.1 -2.1 2.1; 1051 1.6 -2.1 2.2
   30. 1052 1.6 -2.1 2.1; 1053 2.1 -2.1 2.2; 1054 2.1 -2.1 2.1; 1055 2.6 -2.1 2.2
   31. 1056 2.6 -2.1 2.1; 1057 2.6 -2 2.2; 1058 2.1 -2 2.2; 1059 1.6 -2 2.2
   32. 1060 1.1 -2 2.2; 1061 0.6 -2 2.2; 1062 0.6 -2.1 0.1; 1063 0.6 -2.1 0
   33. 1064 1.1 -2.1 0.1; 1065 1.1 -2.1 0; 1066 1.6 -2.1 0.1; 1067 1.6 -2.1 0
   34. 1068 2.1 -2.1 0.1; 1069 2.1 -2.1 0; 1070 2.6 -2.1 0.1; 1071 2.6 -2.1 0
   35. 1072 0.6 -2 0; 1073 1.1 -2 0; 1074 1.6 -2 0; 1075 2.1 -2 0; 1076 2.6 -2 0
   36. 1077 0.6 -1.5 0; 1078 0.1 -1.5 0; 1079 1.1 -1.5 0; 1080 1.6 -1.5 0
   37. 1081 2.1 -1.5 0; 1082 2.6 -1.5 0; 1083 3.1 -1.5 0; 1084 0.6 -1 0
   38. 1085 0.0999999 -1 0; 1086 1.1 -1 0; 1087 1.6 -1 0; 1088 2.1 -1 0
```

39. 1089 2.6 -1 0; 1090 3.1 -1 0; 1091 0.6 -0.5 0; 1092 0.0999999 -0.5 0
40. 1093 1.1 -0.5 0; 1094 1.6 -0.5 0; 1095 2.1 -0.5 0; 1096 2.6 -0.5 0
41. 1097 3.1 -0.5 0; 1098 0.6 0 0; 1099 1.1 0 0; 1100 1.6 0 0; 1101 2.1 0 0
42. 1102 2.6 0 0; 1103 0.6 0.2 0; 1104 1.1 0.2 0; 1105 1.6 0.2 0; 1106 2.1 0.2 0
43. 1107 2.6 0.2 0; 1108 0.6 0.275 0; 1109 1.1 0.275 0; 1110 1.6 0.275 0
44. 1111 2.1 0.275 0; 1112 2.6 0.275 0; 1113 3.2 -1.5 0; 1114 3.2 -1 0
45. 1115 3.2 -0.5 0; 1116 3.2 -1.5 0.1; 1117 3.2 -1 0.1; 1118 3.2 -0.5 0.1
46. 1119 3.2 -1.5 0.6; 1120 3.2 -1.5 1.1; 1121 3.2 -1.5 1.6; 1122 3.2 -1.5 2.1
47. 1123 3.2 -1 0.6; 1124 3.2 -1 1.1; 1125 3.2 -1 1.6; 1126 3.2 -1 2.1
48. 1127 3.2 -0.5 0.6; 1128 3.2 -0.5 1.1; 1129 3.2 -0.5 1.6; 1130 3.2 -0.5 2.1
49. 1131 3.2 0 0.6; 1132 3.2 0 1.1; 1133 3.2 0 1.6; 1134 3.2 0.2 0.6
50. 1135 3.2 0.2 1.1; 1136 3.2 0.2 1.6; 1137 3.2 0.275 0.6; 1138 3.2 0.275 1.1
51. 1139 3.2 0.275 1.6; 1140 3.2 -1.5 2.2; 1141 3.2 -1 2.2; 1142 3.2 -0.5 2.2
52. 1143 3.1 -1.5 2.2; 1144 3.1 -1 2.2; 1145 3.1 -0.5 2.2; 1146 2.6 -1.5 2.2
53. 1147 2.1 -1.5 2.2; 1148 1.6 -1.5 2.2; 1149 1.1 -1.5 2.2; 1150 0.6 -1.5 2.2
54. 1151 0.0999999 -1.5 2.2; 1152 2.6 -1 2.2; 1153 2.1 -1 2.2; 1154 1.6 -1 2.2
55. 1155 1.1 -1 2.2; 1156 0.6 -1 2.2; 1157 0.0999999 -1 2.2; 1158 2.6 -0.5 2.2
56. 1159 2.1 -0.5 2.2; 1160 1.6 -0.5 2.2; 1161 1.1 -0.5 2.2; 1162 0.6 -0.5 2.2
57. 1163 0.1 -0.5 2.2; 1164 2.6 0 2.2; 1165 2.1 0 2.2; 1166 1.6 0 2.2
58. 1167 1.1 0 2.2; 1168 0.6 0 2.2; 1169 2.6 0.2 2.2; 1170 2.1 0.2 2.2
59. 1171 1.6 0.2 2.2; 1172 1.1 0.2 2.2; 1173 0.6 0.2 2.2; 1174 2.6 0.275 2.2
60. 1175 2.1 0.275 2.2; 1176 1.6 0.275 2.2; 1177 1.1 0.275 2.2; 1178 0.6 0.275 2.2
61. 1179 3.1 0.275 1.6; 1180 3.1 0.275 1.1; 1181 3.1 0.275 0.6; 1182 0.6 0.275 2.1
62. 1183 1.1 0.275 2.1; 1184 1.6 0.275 2.1; 1185 2.1 0.275 2.1; 1186 2.6 0.275 2.1
63. 1187 0.1 0.275 1.6; 1188 0 0.275 1.6; 1189 0.1 0.275 1.1; 1190 0 0.275 1.1
64. 1191 0.1 0.275 0.6; 1192 0 0.275 0.6; 1193 0 0.2 1.6; 1194 0 0.2 1.1
65. 1195 0 0.2 0.6; 1196 0 0 1.6; 1197 0 0 1.1; 1198 0 0 0.6; 1199 0 -1.5 0
66. 1200 0 -1 0; 1201 0 -0.5 0; 1202 0 -1.5 0.1; 1203 0 -1 0.1; 1204 0 -0.5 0.1
67. 1205 0 -1.5 2.2; 1206 0 -1 2.2; 1207 0 -0.5 2.2; 1208 0 -1.5 2.1
68. 1209 0 -1 2.1; 1210 0 -0.5 2.1; 1211 0 -1.5 1.6; 1212 0 -1.5 1.1
69. 1213 0 -1.5 0.6; 1214 0 -1 1.6; 1215 0 -1 1.1; 1216 0 -1 0.6; 1217 0 -0.5 1.6
70. 1218 0 -0.5 1.1; 1219 0 -0.5 0.6; 1220 0.6 -2.1 1.6; 1221 1.1 -2.1 1.6
71. 1222 1.6 -2.1 1.6; 1223 2.1 -2.1 1.6; 1224 2.6 -2.1 1.6; 1225 0.6 -2.1 1.1
72. 1226 1.1 -2.1 1.1; 1227 1.6 -2.1 1.1; 1228 2.1 -2.1 1.1; 1229 2.6 -2.1 1.1
73. 1230 0.6 -2.1 0.6; 1231 1.1 -2.1 0.6; 1232 1.6 -2.1 0.6; 1233 2.1 -2.1 0.6
74. 1234 2.6 -2.1 0.6; 1235 0.6 0.275 0.1; 1236 1.1 0.275 0.1; 1237 1.6 0.275 0.1
75. 1238 2.1 0.275 0.1; 1239 2.6 0.275 0.1; 1240 0.6 0.275 1.6; 1241 1.1 0.275 1.6
76. 1242 1.6 0.275 1.6; 1243 2.1 0.275 1.6; 1244 2.6 0.275 1.6; 1245 0.6 0.275 1.1
77. 1246 1.1 0.275 1.1; 1247 1.6 0.275 1.1; 1248 2.1 0.275 1.1; 1249 2.6 0.275 1.1
78. 1250 0.6 0.275 0.6; 1251 1.1 0.275 0.6; 1252 1.6 0.275 0.6; 1253 2.1 0.275 0.6
79. 1254 2.6 0.275 0.6
80. ELEMENT INCIDENCES SHELL
81. 61 1 17 21 9; 62 25 2 10 29; 63 2 34 38 10; 64 42 4 12 46; 65 4 26 30 12
82. 66 18 3 11 22; 67 3 41 45 11; 68 33 1 9 37; 69 6 36 40 14; 70 27 6 14 31
83. 71 44 8 16 48; 72 8 28 32 16; 73 20 7 15 24; 74 7 43 47 15; 75 5 19 23 13
84. 76 35 5 13 39; 977 54 36 6 27; 978 35 50 19 5; 979 7 20 58 43; 980 28 8 44 62
85. 1181 10 38 952 944; 1182 29 10 944 949; 1183 46 12 946 954; 1184 12 30 950 946
86. 1185 22 11 945 948; 1186 11 45 953 945; 1187 9 21 947 943; 1188 37 9 943 951
87. 1240 1014 952 944 949; 1241 951 1013 947 943; 1242 945 948 1015 953
88. 1243 950 946 954 1016; 1467 54 1029 1030 36; 1468 1029 1031 1032 1030
89. 1469 1031 1033 1034 1032; 1470 1033 62 44 1034; 1471 36 1030 1035 40
90. 1472 1030 1032 1036 1035; 1473 1032 1034 1037 1036; 1474 1034 44 48 1037
91. 1475 58 1038 1039 43; 1476 1038 1040 1041 1039; 1477 1040 1042 1043 1041
92. 1478 1042 50 35 1043; 1479 43 1039 1044 47; 1480 1039 1041 1045 1044
93. 1481 1041 1043 1046 1045; 1482 1043 35 39 1046; 1483 20 1047 1048 58
94. 1484 1047 1049 1050 1048; 1485 1049 1051 1052 1050; 1486 1051 1053 1054 1052

95. 1487 1053 1055 1056 1054; 1488 1055 28 62 1056; 1489 28 1055 1057 32
96. 1490 1055 1053 1058 1057; 1491 1053 1051 1059 1058; 1492 1051 1049 1060 1059
97. 1493 1049 1047 1061 1060; 1494 1047 20 24 1061; 1495 50 1062 1063 19
98. 1496 1062 1064 1065 1063; 1497 1064 1066 1067 1065; 1498 1066 1068 1069 1067
99. 1499 1068 1070 1071 1069; 1500 1070 54 27 1071; 1501 19 1063 1072 23
100. 1502 1063 1065 1073 1072; 1503 1065 1067 1074 1073; 1504 1067 1069 1075 1074
101. 1505 1069 1071 1076 1075; 1506 1071 27 31 1076; 1507 23 1072 1077 1078
102. 1508 1072 1073 1079 1077; 1509 1073 1074 1080 1079; 1510 1074 1075 1081 1080
103. 1511 1075 1076 1082 1081; 1512 1076 31 1083 1082; 1513 1078 1077 1084 1085
104. 1514 1077 1079 1086 1084; 1515 1079 1080 1087 1086; 1516 1080 1081 1088 1087
105. 1517 1081 1082 1089 1088; 1518 1082 1083 1090 1089; 1519 1085 1084 1091 1092
106. 1520 1084 1086 1093 1091; 1521 1086 1087 1094 1093; 1522 1087 1088 1095 1094
107. 1523 1088 1089 1096 1095; 1524 1089 1090 1097 1096; 1525 1092 1091 1098 17
108. 1526 1091 1093 1099 1098; 1527 1093 1094 1100 1099; 1528 1094 1095 1101 1100
109. 1529 1095 1096 1102 1101; 1530 1096 1097 25 1102; 1531 17 1098 1103 21
110. 1532 1098 1099 1104 1103; 1533 1099 1100 1105 1104; 1534 1100 1101 1106 1105
111. 1535 1101 1102 1107 1106; 1536 1102 25 29 1107; 1537 21 1103 1108 947
112. 1538 1103 1104 1109 1108; 1539 1104 1105 1110 1109; 1540 1105 1106 1111 1110
113. 1541 1106 1107 1112 1111; 1542 1107 29 949 1112; 1543 31 14 1113 1083
114. 1544 1083 1113 1114 1090; 1545 1090 1114 1115 1097; 1546 1097 1115 2 25
115. 1547 14 40 1116 1113; 1548 1113 1116 1117 1114; 1549 1114 1117 1118 1115
116. 1550 1115 1118 34 2; 1551 40 1035 1119 1116; 1552 1035 1036 1120 1119
117. 1553 1036 1037 1121 1120; 1554 1037 48 1122 1121; 1555 1116 1119 1123 1117
118. 1556 1119 1120 1124 1123; 1557 1120 1121 1125 1124; 1558 1121 1122 1126 1125
119. 1559 1117 1123 1127 1118; 1560 1123 1124 1128 1127; 1561 1124 1125 1129 1128
120. 1562 1125 1126 1130 1129; 1563 1118 1127 1131 34; 1564 1127 1128 1132 1131
121. 1565 1128 1129 1133 1132; 1566 1129 1130 42 1133; 1567 34 1131 1134 38
122. 1568 1131 1132 1135 1134; 1569 1132 1133 1136 1135; 1570 1133 42 46 1136
123. 1571 38 1134 1137 952; 1572 1134 1135 1138 1137; 1573 1135 1136 1139 1138
124. 1574 1136 46 954 1139; 1575 48 16 1140 1122; 1576 1122 1140 1141 1126
125. 1577 1126 1141 1142 1130; 1578 1130 1142 4 42; 1579 16 32 1143 1140
126. 1580 1140 1143 1144 1141; 1581 1141 1144 1145 1142; 1582 1142 1145 26 4
127. 1583 32 1057 1146 1143; 1584 1057 1058 1147 1146; 1585 1058 1059 1148 1147
128. 1586 1059 1060 1149 1148; 1587 1060 1061 1150 1149; 1588 1061 24 1151 1150
129. 1589 1143 1146 1152 1144; 1590 1146 1147 1153 1152; 1591 1147 1148 1154 1153
130. 1592 1148 1149 1155 1154; 1593 1149 1150 1156 1155; 1594 1150 1151 1157 1156
131. 1595 1144 1152 1158 1145; 1596 1152 1153 1159 1158; 1597 1153 1154 1160 1159
132. 1598 1154 1155 1161 1160; 1599 1155 1156 1162 1161; 1600 1156 1157 1163 1162
133. 1601 1145 1158 1164 26; 1602 1158 1159 1165 1164; 1603 1159 1160 1166 1165
134. 1604 1160 1161 1167 1166; 1605 1161 1162 1168 1167; 1606 1162 1163 18 1168
135. 1607 26 1164 1169 30; 1608 1164 1165 1170 1169; 1609 1165 1166 1171 1170
136. 1610 1166 1167 1172 1171; 1611 1167 1168 1173 1172; 1612 1168 18 22 1173
137. 1613 30 1169 1174 950; 1614 1169 1170 1175 1174; 1615 1170 1171 1176 1175
138. 1616 1171 1172 1177 1176; 1617 1172 1173 1178 1177; 1618 1173 22 948 1178
139. 1619 1016 954 1139 1179; 1620 1179 1139 1138 1180; 1621 1180 1138 1137 1181
140. 1622 1181 1137 952 1014; 1623 948 1178 1182 1015; 1624 1178 1177 1183 1182
141. 1625 1177 1176 1184 1183; 1626 1176 1175 1185 1184; 1627 1175 1174 1186 1185
142. 1628 1174 950 1016 1186; 1629 953 1015 1187 1188; 1630 1188 1187 1189 1190
143. 1631 1190 1189 1191 1192; 1632 1192 1191 1013 951; 1633 45 1193 1188 953
144. 1634 1193 1194 1190 1188; 1635 1194 1195 1192 1190; 1636 1195 37 951 1192
145. 1637 41 1196 1193 45; 1638 1196 1197 1194 1193; 1639 1197 1198 1195 1194
146. 1640 1198 33 37 1195; 1641 13 23 1078 1199; 1642 1199 1078 1085 1200
147. 1643 1200 1085 1092 1201; 1644 1201 1092 17 1; 1645 39 13 1199 1202
148. 1646 1202 1199 1200 1203; 1647 1203 1200 1201 1204; 1648 1204 1201 1 33
149. 1649 24 15 1205 1151; 1650 1151 1205 1206 1157; 1651 1157 1206 1207 1163
150. 1652 1163 1207 3 18; 1653 15 47 1208 1205; 1654 1205 1208 1209 1206

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151. 1655 1206 1209 1210 1207; 1656 1207 1210 41 3; 1657 47 1044 1211 1208
152. 1658 1044 1045 1212 1211; 1659 1045 1046 1213 1212; 1660 1046 39 1202 1213
153. 1661 1208 1211 1214 1209; 1662 1211 1212 1215 1214; 1663 1212 1213 1216 1215
154. 1664 1213 1202 1203 1216; 1665 1209 1214 1217 1210; 1666 1214 1215 1218 1217
155. 1667 1215 1216 1219 1218; 1668 1216 1203 1204 1219; 1669 1210 1217 1196 41
156. 1670 1217 1218 1197 1196; 1671 1218 1219 1198 1197; 1672 1219 1204 33 1198
157. 1673 58 1048 1220 1038; 1674 1048 1050 1221 1220; 1675 1050 1052 1222 1221
158. 1676 1052 1054 1223 1222; 1677 1054 1056 1224 1223; 1678 1056 62 1033 1224
159. 1679 1038 1220 1225 1040; 1680 1220 1221 1226 1225; 1681 1221 1222 1227 1226
160. 1682 1222 1223 1228 1227; 1683 1223 1224 1229 1228; 1684 1224 1033 1031 1229
161. 1685 1040 1225 1230 1042; 1686 1225 1226 1231 1230; 1687 1226 1227 1232 1231
162. 1688 1227 1228 1233 1232; 1689 1228 1229 1234 1233; 1690 1229 1031 1029 1234
163. 1691 1042 1230 1062 50; 1692 1230 1231 1064 1062; 1693 1231 1232 1066 1064
164. 1694 1232 1233 1068 1066; 1695 1233 1234 1070 1068; 1696 1234 1029 54 1070
165. 1697 1013 1235 1108 947; 1698 1235 1236 1109 1108; 1699 1236 1237 1110 1109
166. 1700 1237 1238 1111 1110; 1701 1238 1239 1112 1111; 1702 1239 1014 949 1112
167. 1703 1015 1182 1240 1187; 1704 1182 1183 1241 1240; 1705 1183 1184 1242 1241
168. 1706 1184 1185 1243 1242; 1707 1185 1186 1244 1243; 1708 1186 1016 1179 1244
169. 1709 1187 1240 1245 1189; 1710 1240 1241 1246 1245; 1711 1241 1242 1247 1246
170. 1712 1242 1243 1248 1247; 1713 1243 1244 1249 1248; 1714 1244 1179 1180 1249
171. 1715 1189 1245 1250 1191; 1716 1245 1246 1251 1250; 1717 1246 1247 1252 1251
172. 1718 1247 1248 1253 1252; 1719 1248 1249 1254 1253; 1720 1249 1180 1181 1254
173. 1721 1191 1250 1235 1013; 1722 1250 1251 1236 1235; 1723 1251 1252 1237 1236
174. 1724 1252 1253 1238 1237; 1725 1253 1254 1239 1238; 1726 1254 1181 1014 1239
175. START GROUP DEFINITION
176. ELEMENT
177. _WALL 1507 TO 1536 1551 TO 1570 1583 TO 1612 1637 TO 1640 1657 TO 1672
178. _TOP_SLAB 1703 TO 1726
179. _BOT_SLAB 1673 TO 1696
180. END GROUP DEFINITION
181. ELEMENT PROPERTY
182. 61 TO 76 1181 TO 1188 1471 TO 1474 1479 TO 1482 1489 TO 1494 1501 TO 1618 -
183. 1633 TO 1672 THICKNESS 0.2
184. 977 TO 980 1467 TO 1470 1475 TO 1478 1483 TO 1488 1495 TO 1500 1673 TO 1695 -
185. 1696 THICKNESS 0.2
186. 1240 TO 1243 1619 TO 1632 1697 TO 1726 THICKNESS 0.15
187. DEFINE MATERIAL START
188. ISOTROPIC CONCRETE
189. E 2.17185E+007
190. POISSON 0.17
191. DENSITY 25
192. ALPHA 1E-005
193. DAMP 0.05
194. TYPE CONCRETE
195. STRENGTH FCU 35000
196. END DEFINE MATERIAL
197. CONSTANTS
198. MATERIAL CONCRETE ALL
199. SUPPORTS
200. 977 TO 980 1467 TO 1470 1475 TO 1478 1483 TO 1488 1495 TO 1500 1673 TO 1695 -
201. 1696 PLATE MAT DIRECT Y SUBGRADE 102960 PRINT
```

JOINT	INFLUENCE AREA
5	0.002500

6	0.002500
7	0.002500
8	0.002500
19	0.015000
20	0.015000
27	0.015000
28	0.015000
35	0.015000
36	0.015000
43	0.015000
44	0.015000
50	0.090000
54	0.090000
58	0.090000
62	0.090000
1029	0.150000
1030	0.025000
1031	0.150000
1032	0.025000
1033	0.150000
1034	0.025000
1038	0.150000
1039	0.025000
1040	0.150000
1041	0.025000
1042	0.150000
1043	0.025000
1047	0.025000
1048	0.150000
1049	0.025000
1050	0.150000
1051	0.025000
1052	0.150000
1053	0.025000
1054	0.150000
1055	0.025000
1056	0.150000
1062	0.150000
1063	0.025000
1064	0.150000
1065	0.025000
1066	0.150000
1067	0.025000
1068	0.150000
1069	0.025000
1070	0.150000
1071	0.025000
1220	0.250000
1221	0.250000
1222	0.250000
1223	0.250000
1224	0.250000
1225	0.250000
1226	0.250000
1227	0.250000
1228	0.250000

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      1229          0.250000
      1230          0.250000
      1231          0.250000
      1232          0.250000
      1233          0.250000
      1234          0.250000
     TOTAL          7.040000
202. LOAD 1 LOADTYPE DEAD  TITLE DEAD LOAD
203. SELFWEIGHT Y -1
204. ELEMENT LOAD
205. 1673 TO 1696 PR -1.2
206. LOAD 2 LOADTYPE LIVE  TITLE LIVE LOAD
207. ELEMENT LOAD
208. 1703 TO 1726 PR -5
209. LOAD 3 LOADTYPE FLUIDS  TITLE WATER PRESSURE
210. ELEMENT LOAD
211. 1507 TO 1512 1551 TO 1554 1583 TO 1588 1657 TO 1660 TRAP JT -22 -22 -17 -17
212. 1513 TO 1518 1555 TO 1558 1589 TO 1594 1661 TO 1664 TRAP JT -17 -17 -12 -12
213. 1519 TO 1524 1559 TO 1562 1595 TO 1600 1665 TO 1668 TRAP JT -12 -12 -7 -7
214. 1525 TO 1530 1563 TO 1566 1601 TO 1606 1669 TO 1672 TRAP JT -7 -7 -2 -2
215. 1531 TO 1536 1567 TO 1570 1607 TO 1612 1637 TO 1639 -
216. 1640 TRAP JT -2 -2 -4.47035E-007 -4.47035E-007
217. 1673 TO 1696 PR -22
218. LOAD 4 LOADTYPE SOIL  TITLE SOIL PRESSURE
219. ELEMENT LOAD
220. 1507 TO 1512 1543 1547 1551 TO 1554 1575 1579 1583 TO 1588 1641 1645 1649 -
221. 1653 1657 TO 1660 TRAP JT 18 18 13.5 13.5
222. 1513 TO 1518 1544 1548 1555 TO 1558 1576 1580 1589 TO 1594 1642 1646 1650 -
223. 1654 1661 TO 1664 TRAP JT 13.5 13.5 9 9
224. 1519 TO 1524 1545 1549 1559 TO 1562 1577 1581 1595 TO 1600 1643 1647 1651 -
225. 1655 1665 TO 1668 TRAP JT 9 9 4.5 4.5
226. 1525 TO 1530 1546 1550 1563 TO 1566 1578 1582 1601 TO 1606 1644 1648 1652 -
227. 1656 1669 TO 1672 TRAP JT 4.5 4.5 0 0
228. 1507 TO 1530 1543 TO 1566 1575 TO 1606 1641 TO 1672 PR 5
229. ******FACTORED  LOAD COMBINATIONS*****
230. LOAD COMB 101 1.4*(D+F)
231. 1 1.4 3 1.4
232. LOAD COMB 102 1.2*(D+F)+1.6*(L+H)
233. 1 1.2 3 1.2 2 1.6 4 1.6
234. LOAD COMB 103 1.4*(D+H)
235. 1 1.4 4 1.4
236. ******UNFACTORED  LOAD COMBINATIONS*****
237. LOAD COMB 201 1.0*(D+F)
238. 1 1.0 3 1.0
239. LOAD COMB 202 1.0*(D+F)+1.0*(L+H)
240. 1 1.0 3 1.0 2 1.0 4 1.0
241. LOAD COMB 203 1.0*(D+H)
242. 1 1.0 4 1.0
243. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**294**	**NUMBER OF MEMBERS**	**0**
NUMBER OF PLATES	**292**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**63**

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 1575

244. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 15,2020 TIME= 10:35:10 ****

ATTACHMENT 3 STAAD INPUT FOR SUMP TANK

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08-Oct-20
ENGINEER NAME BJB
CHECKER NAME SJM
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 3.3 0 0; 3 0 0 1.2; 4 3.3 0 1.2; 5 0 -2.1 0; 6 3.3 -2.1 0;
7 0 -2.1 1.2; 8 3.3 -2.1 1.2; 13 0 -2 0; 14 3.3 -2 0; 15 0 -2 1.2;
16 3.3 -2 1.2; 17 0.1 0 0; 18 0.1 0 1.2; 19 0.1 -2.1 0; 20 0.1 -2.1 1.2;
23 0.1 -2 0; 24 0.1 -2 1.2; 25 3.2 0 0; 26 3.2 0 1.2; 27 3.2 -2.1 0;
28 3.2 -2.1 1.2; 31 3.2 -2 0; 32 3.2 -2 1.2; 33 0 0 0.1; 34 3.3 0 0.1;
35 0 -2.1 0.1; 36 3.3 -2.1 0.1; 39 0 -2 0.1; 40 3.3 -2 0.1; 41 0 0 1.1;
42 3.3 0 1.1; 43 0 -2.1 1.1; 44 3.3 -2.1 1.1; 47 0 -2 1.1; 48 3.3 -2 1.1;
50 0.1 -2.1 0.1; 54 3.2 -2.1 0.1; 58 0.1 -2.1 1.1; 62 3.2 -2.1 1.1;
943 0 0.5 0; 944 3.3 0.5 0; 945 0 0.5 1.2; 946 3.3 0.5 1.2; 947 0.1 0.5 0;
948 0.1 0.5 1.2; 949 3.2 0.5 0; 950 3.2 0.5 1.2; 951 0 0.5 0.1;
952 3.3 0.5 0.1; 953 0 0.5 1.1; 954 3.3 0.5 1.1; 1029 3.2 -2.1 0.6;
1030 3.3 -2.1 0.6; 1035 3.3 -2 0.6; 1038 0.1 -2.1 0.6; 1039 0 -2.1 0.6;
1044 0 -2 0.6; 1047 0.6 -2.1 1.2; 1048 0.6 -2.1 1.1; 1049 1.1 -2.1 1.2;
1050 1.1 -2.1 1.1; 1051 1.6 -2.1 1.2; 1052 1.6 -2.1 1.1; 1053 2.1 -2.1 1.2;
1054 2.1 -2.1 1.1; 1055 2.75 -2.1 1.2; 1056 2.75 -2.1 1.1; 1057 2.75 -2 1.2;
1058 2.1 -2 1.2; 1059 1.6 -2 1.2; 1060 1.1 -2 1.2; 1061 0.6 -2 1.2;
1062 0.6 -2.1 0.1; 1063 0.6 -2.1 0; 1064 1.1 -2.1 0.1; 1065 1.1 -2.1 0;
1066 1.6 -2.1 0.1; 1067 1.6 -2.1 0; 1068 2.1 -2.1 0.1; 1069 2.1 -2.1 0;
1070 2.75 -2.1 0.1; 1071 2.75 -2.1 0; 1072 0.6 -2 0; 1073 1.1 -2 0;
1074 1.6 -2 0; 1075 2.1 -2 0; 1076 2.75 -2 0; 1077 0.6 -1.5 0; 1078 0.1 -1.5 0;
1079 1.1 -1.5 0; 1080 1.6 -1.5 0; 1081 2.1 -1.5 0; 1082 2.75 -1.5 0;
1083 3.2 -1.5 0; 1084 0.6 -1 0; 1085 0.0999999 -1 0; 1086 1.1 -1 0;
1087 1.6 -1 0; 1088 2.1 -1 0; 1089 2.75 -1 0; 1090 3.2 -1 0; 1091 0.6 -0.5 0;
1092 0.0999999 -0.5 0; 1093 1.1 -0.5 0; 1094 1.6 -0.5 0; 1095 2.1 -0.5 0;
1096 2.75 -0.5 0; 1097 3.2 -0.5 0; 1098 0.6 0 0; 1099 1.1 0 0; 1100 1.6 0 0;
1101 2.1 0 0; 1102 2.75 0 0; 1108 0.6 0.5 0; 1109 1.1 0.5 0; 1110 1.6 0.5 0;
1111 2.1 0.5 0; 1112 2.75 0.5 0; 1113 3.3 -1.5 0; 1114 3.3 -1 0;
1115 3.3 -0.5 0; 1116 3.3 -1.5 0.1; 1117 3.3 -1 0.1; 1118 3.3 -0.5 0.1;
1119 3.3 -1.5 0.6; 1122 3.3 -1.5 1.1; 1123 3.3 -1 0.6; 1126 3.3 -1 1.1;
1127 3.3 -0.5 0.6; 1130 3.3 -0.5 1.1; 1131 3.3 0 0.6; 1137 3.3 0.5 0.6;
1140 3.3 -1.5 1.2; 1141 3.3 -1 1.2; 1142 3.3 -0.5 1.2; 1143 3.2 -1.5 1.2;
1144 3.2 -1 1.2; 1145 3.2 -0.5 1.2; 1146 2.75 -1.5 1.2; 1147 2.1 -1.5 1.2;
1148 1.6 -1.5 1.2; 1149 1.1 -1.5 1.2; 1150 0.6 -1.5 1.2;
1151 0.0999999 -1.5 1.2; 1152 2.75 -1 1.2; 1153 2.1 -1 1.2; 1154 1.6 -1 1.2;
1155 1.1 -1 1.2; 1156 0.6 -1 1.2; 1157 0.0999999 -1 1.2; 1158 2.75 -0.5 1.2;
1159 2.1 -0.5 1.2; 1160 1.6 -0.5 1.2; 1161 1.1 -0.5 1.2; 1162 0.6 -0.5 1.2;
1163 0.1 -0.5 1.2; 1164 2.75 0 1.2; 1165 2.1 0 1.2; 1166 1.6 0 1.2;
1167 1.1 0 1.2; 1168 0.6 0 1.2; 1174 2.75 0.5 1.2; 1175 2.1 0.5 1.2;
1176 1.6 0.5 1.2; 1177 1.1 0.5 1.2; 1178 0.6 0.5 1.2; 1188 0 0.5 0.6;
1193 0 0 0.6; 1199 0 -1.5 0; 1200 0 -1 0; 1201 0 -0.5 0; 1202 0 -1.5 0.1;
1203 0 -1 0.1; 1204 0 -0.5 0.1; 1205 0 -1.5 1.2; 1206 0 -1 1.2;
1207 0 -0.5 1.2; 1208 0 -1.5 1.1; 1209 0 -1 1.1; 1210 0 -0.5 1.1;
1211 0 -1.5 0.6; 1214 0 -1 0.6; 1217 0 -0.5 0.6; 1220 0.6 -2.1 0.6;
1221 1.1 -2.1 0.6; 1222 1.6 -2.1 0.6; 1223 2.1 -2.1 0.6; 1224 2.75 -2.1 0.6;
1235 2.20333 -2.1 1.2; 1236 2.20333 -2.1 1.1; 1237 2.30667 -2.1 1.2;
1238 2.30667 -2.1 1.1; 1239 2.30667 -2 1.2; 1240 2.20333 -2 1.2;
1241 2.20333 -2.1 0.1; 1242 2.20333 -2.1 0; 1243 2.30667 -2.1 0.1;
1244 2.30667 -2.1 0; 1245 2.20333 -2 0; 1246 2.30667 -2 0; 1247 2.20333 -1.5 0;
1248 2.30667 -1.5 0; 1249 2.20333 -1 0; 1250 2.30667 -1 0; 1251 2.20333 -0.5 0;
1252 2.30667 -0.5 0; 1253 2.20333 0 0; 1254 2.30667 0 0; 1257 2.20333 0.5 0;
1258 2.30667 0.5 0; 1259 2.30667 -1.5 1.2; 1260 2.20333 -1.5 1.2;
1261 2.30667 -1 1.2; 1262 2.20333 -1 1.2; 1263 2.30667 -0.5 1.2;
1264 2.20333 -0.5 1.2; 1265 2.30667 0 1.2; 1266 2.20333 0 1.2;
1269 2.30667 0.5 1.2; 1270 2.20333 0.5 1.2; 1271 2.20333 -2.1 0.6;
1272 2.30667 -2.1 0.6; 1277 2.20333 -2 1.1; 1278 2.20333 -2 0.1;
1279 2.20333 -2 0.6; 1282 2.20333 -1.5 0.1; 1283 2.20333 -1 0.1;
1284 2.20333 -0.5 0.1; 1285 2.20333 0 0.1; 1287 2.20333 0.5 0.1;
1288 2.20333 -1.5 1.1; 1289 2.20333 -1 1.1; 1290 2.20333 -0.5 1.1;
1291 2.20333 0 1.1; 1293 2.20333 0.5 1.1; 1294 2.20333 -1.5 0.6;
1297 2.20333 -1 0.6; 1300 2.20333 -0.5 0.6; 1303 2.20333 0 0.6;
1309 2.20333 0.5 0.6;
MEMBER INCIDENCES
1799 943 947; 1800 947 1108; 1801 1108 1109; 1802 1109 1110; 1803 1110 1111;
1804 1111 1257; 1805 1257 1258; 1806 1258 1112; 1807 1112 949; 1808 949 944;
1809 945 948; 1810 948 1178; 1811 1178 1177; 1812 1177 1176; 1813 1176 1175;
1814 1175 1270; 1815 1270 1269; 1816 1269 1174; 1817 1174 950; 1818 950 946;
1819 943 951; 1820 951 1188; 1821 1188 953; 1822 953 945; 1823 944 952;
1824 952 1137; 1825 1137 954; 1826 954 946; 1827 1257 1287; 1828 1287 1309;
1829 1309 1293; 1830 1293 1270;
ELEMENT INCIDENCES SHELL
69 6 36 40 14; 70 27 6 14 31; 71 44 8 16 48; 72 8 28 32 16; 73 20 7 15 24;
74 7 43 47 15; 75 5 19 23 13; 76 35 5 13 39; 977 54 36 6 27; 978 35 50 19 5;
979 7 20 58 43; 980 28 8 44 62; 1181 2 34 952 944; 1182 25 2 944 949;
1183 42 4 946 954; 1184 4 26 950 946; 1185 18 3 945 948; 1186 3 41 953 945;
1187 1 17 947 943; 1188 33 1 943 951; 1467 54 1029 1030 36;
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1470 1029 62 44 1030; 1471 36 1030 1035 40; 1474 1030 44 48 1035;
1475 58 1038 1039 43; 1478 1038 50 35 1039; 1479 43 1039 1044 47;
1482 1039 35 39 1044; 1483 20 1047 1048 58; 1484 1047 1049 1050 1048;
1485 1049 1051 1052 1050; 1486 1051 1053 1054 1052; 1488 1055 28 62 1056;
1489 28 1055 1057 32; 1491 1053 1051 1059 1058; 1492 1051 1049 1060 1059;
1493 1049 1047 1061 1060; 1494 1047 20 24 1061; 1495 50 1062 1063 19;
1496 1062 1064 1065 1063; 1497 1064 1066 1067 1065; 1498 1066 1068 1069 1067;
1500 1070 54 27 1071; 1501 19 1063 1072 23; 1502 1063 1065 1073 1072;
1503 1065 1067 1074 1073; 1504 1067 1069 1075 1074; 1506 1071 27 31 1076;
1507 23 1072 1077 1078; 1508 1072 1073 1079 1077; 1509 1073 1074 1080 1079;
1510 1074 1075 1081 1080; 1512 1076 31 1083 1082; 1513 1078 1077 1084 1085;
1514 1077 1079 1086 1084; 1515 1079 1080 1087 1086; 1516 1080 1081 1088 1087;
1518 1082 1083 1090 1089; 1519 1085 1084 1091 1092; 1520 1084 1086 1093 1091;
1521 1086 1087 1094 1093; 1522 1087 1088 1095 1094; 1524 1089 1090 1097 1096;
1525 1092 1091 1098 17; 1526 1091 1093 1099 1098; 1527 1093 1094 1100 1099;
1528 1094 1095 1101 1100; 1530 1096 1097 25 1102; 1537 17 1098 1108 947;
1538 1098 1099 1109 1108; 1539 1099 1100 1110 1109; 1540 1100 1101 1111 1110;
1542 1102 25 949 1112; 1543 31 14 1113 1083; 1544 1083 1113 1114 1090;
1545 1090 1114 1115 1097; 1546 1097 1115 2 25; 1547 14 40 1116 1113;
1548 1113 1116 1117 1114; 1549 1114 1117 1118 1115; 1550 1115 1118 34 2;
1551 40 1035 1119 1116; 1554 1035 48 1122 1119; 1555 1116 1119 1123 1117;
1558 1119 1122 1126 1123; 1559 1117 1123 1127 1118; 1562 1123 1126 1130 1127;
1563 1118 1127 1131 34; 1566 1127 1130 42 1131; 1571 34 1131 1137 952;
1574 1131 42 954 1137; 1575 48 16 1140 1122; 1576 1122 1140 1141 1126;
1577 1126 1141 1142 1130; 1578 1130 1142 4 42; 1579 16 32 1143 1140;
1580 1140 1143 1144 1141; 1581 1141 1144 1145 1142; 1582 1142 1145 26 4;
1583 32 1057 1146 1143; 1585 1058 1059 1148 1147; 1586 1059 1060 1149 1148;
1587 1060 1061 1150 1149; 1588 1061 24 1151 1150; 1589 1143 1146 1152 1144;
1591 1147 1148 1154 1153; 1592 1148 1149 1155 1154; 1593 1149 1150 1156 1155;
1594 1150 1151 1157 1156; 1595 1144 1152 1158 1145; 1597 1153 1154 1160 1159;
1598 1154 1155 1161 1160; 1599 1155 1156 1162 1161; 1600 1156 1157 1163 1162;
1601 1145 1158 1164 26; 1603 1159 1160 1166 1165; 1604 1160 1161 1167 1166;
1605 1161 1162 1168 1167; 1606 1162 1163 18 1168; 1613 26 1164 1174 950;
1615 1165 1166 1176 1175; 1616 1166 1167 1177 1176; 1617 1167 1168 1178 1177;
1618 1168 18 948 1178; 1633 41 1193 1188 953; 1636 1193 33 951 1188;
1641 13 23 1078 1199; 1642 1199 1078 1085 1200; 1643 1200 1085 1092 1201;
1644 1201 1092 17 1; 1645 39 13 1199 1202; 1646 1202 1199 1200 1203;
1647 1203 1200 1201 1204; 1648 1204 1201 1 33; 1649 24 15 1205 1151;
1650 1151 1205 1206 1157; 1651 1157 1206 1207 1163; 1652 1163 1207 3 18;
1653 15 47 1208 1205; 1654 1205 1208 1209 1206; 1655 1206 1209 1210 1207;
1656 1207 1210 41 3; 1657 47 1044 1211 1208; 1660 1044 39 1202 1211;
1661 1208 1211 1214 1209; 1664 1211 1202 1203 1214; 1665 1209 1214 1217 1210;
1668 1214 1203 1204 1217; 1669 1210 1217 1193 41; 1672 1217 1204 33 1193;
1673 58 1048 1220 1038; 1674 1048 1050 1221 1220; 1675 1050 1052 1222 1221;
1676 1052 1054 1223 1222; 1678 1056 62 1029 1224; 1691 1038 1220 1062 50;
1692 1220 1221 1064 1062; 1693 1221 1222 1066 1064; 1694 1222 1223 1068 1066;
1696 1224 1029 54 1070; 1697 1053 1235 1236 1054; 1698 1235 1237 1238 1236;
1699 1237 1055 1056 1238; 1700 1055 1237 1239 1057; 1701 1237 1235 1240 1239;
1702 1235 1053 1058 1240; 1703 1068 1241 1242 1069; 1704 1241 1243 1244 1242;
1705 1243 1070 1071 1244; 1706 1069 1242 1245 1075; 1707 1242 1244 1246 1245;
1708 1244 1071 1076 1246; 1709 1075 1245 1247 1081; 1710 1245 1246 1248 1247;
1711 1246 1076 1082 1248; 1712 1081 1247 1249 1088; 1713 1247 1248 1250 1249;
1714 1248 1082 1089 1250; 1715 1088 1249 1251 1095; 1716 1249 1250 1252 1251;
1717 1250 1089 1096 1252; 1718 1095 1251 1253 1101; 1719 1251 1252 1254 1253;
1720 1252 1096 1102 1254; 1724 1101 1253 1257 1111; 1725 1253 1254 1258 1257;
1726 1254 1102 1112 1258; 1727 1057 1239 1259 1146; 1728 1239 1240 1260 1259;
1729 1240 1058 1147 1260; 1730 1146 1259 1261 1152; 1731 1259 1260 1262 1261;
1732 1260 1147 1153 1262; 1733 1152 1261 1263 1158; 1734 1261 1262 1264 1263;
1735 1262 1153 1159 1264; 1736 1158 1263 1265 1164; 1737 1263 1264 1266 1265;
1738 1264 1159 1165 1266; 1742 1164 1265 1269 1174; 1743 1265 1266 1270 1269;
1744 1266 1165 1175 1270; 1745 1054 1236 1271 1223; 1746 1236 1238 1272 1271;
1747 1238 1056 1224 1272; 1754 1223 1271 1241 1068; 1755 1271 1272 1243 1241;
1756 1272 1224 1070 1243; 1757 1242 1241 1278 1245; 1758 1236 1235 1240 1277;
1759 1241 1271 1279 1278; 1762 1271 1236 1277 1279; 1763 1245 1278 1282 1247;
1764 1247 1282 1283 1249; 1765 1249 1283 1284 1251; 1766 1251 1284 1285 1253;
1768 1253 1285 1287 1257; 1769 1277 1240 1260 1288; 1770 1288 1260 1262 1289;
1771 1289 1262 1264 1290; 1772 1290 1264 1266 1291; 1774 1291 1266 1270 1293;
1775 1278 1279 1294 1282; 1778 1279 1277 1288 1294; 1779 1282 1294 1297 1283;
1782 1294 1288 1289 1297; 1783 1283 1297 1300 1284; 1786 1297 1289 1290 1300;
1787 1284 1300 1303 1285; 1790 1300 1290 1291 1303; 1795 1285 1303 1309 1287;
1798 1303 1291 1293 1309;
START GROUP DEFINITION
ELEMENT
_WALL 1507 TO 1510 1512 TO 1516 1518 TO 1522 1524 TO 1528 1530 1537 TO 1540 -
1542 1551 1554 1555 1558 1559 1562 1563 1566 1571 1574 1583 1585 TO 1589 -
1591 TO 1595 1597 TO 1601 1603 TO 1606 1613 1615 TO 1618 1633 1636 1657 1660 -
1661 1664 1665 1668 1669 1672 1711 1714 1717 1720 1726 1727 1730 1733 1736 -
1742
_BOT_SLAB 1673 TO 1676 1678 1691 TO 1694 1696 1747 1756
_BUFFER_WALL 1775 1778 1779 1782 1783 1786 1787 1790 1795 1798
MEMBER
FLOOR
_GRATING 1799 TO 1830
END GROUP DEFINITION
```

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ELEMENT PROPERTY
69 TO 76 1181 TO 1188 1471 1474 1479 1482 1489 1491 TO 1494 1501 TO 1504 1506 -
1507 TO 1510 1512 TO 1516 1518 TO 1522 1524 TO 1528 1530 1537 TO 1540 1542 -
1543 TO 1551 1554 1555 1558 1559 1562 1563 1566 1571 1574 TO 1583 -
1585 TO 1589 1591 TO 1595 1597 TO 1601 1603 TO 1606 1613 1615 TO 1618 1633 -
1636 1641 TO 1657 1660 1661 1664 1665 1668 1669 1672 1700 TO 1702 -
1706 TO 1720 1724 TO 1738 1742 TO 1744 THICKNESS 0.2
977 TO 980 1467 1470 1475 1478 1483 TO 1486 1488 1495 TO 1498 1500 -
1673 TO 1676 1678 1691 TO 1694 1696 TO 1699 1703 TO 1705 1745 TO 1747 1754 -
1755 TO 1756 THICKNESS 0.2
1757 TO 1759 1762 TO 1766 1768 TO 1772 1774 1775 1778 1779 1782 1783 1786 -
1787 1790 1795 1798 THICKNESS 0.2
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 35000
ISOTROPIC DENSITYZERO
E 2.17185e+007
POISSON 0.17
DENSITY 0
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 0
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1799 TO 1830 PRIS YD 0.05 ZD 0.05
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
977 TO 980 1467 1470 1475 1478 1483 TO 1486 1488 1495 TO 1498 1500 -
1673 TO 1676 1678 1691 TO 1694 1696 TO 1699 1703 TO 1705 1745 TO 1747 1754 -
1755 TO 1756 PLATE MAT DIRECT Y SUBGRADE 102960 PRINT
LOAD 1 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
1673 TO 1676 1678 1691 TO 1694 1696 1747 1756 PR -1.2
FLOOR LOAD
_GRATING FLOAD -0.05 GY
LOAD 2 LOADTYPE Live  TITLE LIVE LOAD
FLOOR LOAD
_GRATING FLOAD -5 GY
LOAD 3 LOADTYPE Fluids  TITLE WATER PRESSURE1
ELEMENT LOAD
1673 TO 1676 1691 TO 1694 PR -25
1507 TO 1510 1585 TO 1588 1657 1660 1775 1778 TRAP JT -25 -25 -20 -20
1513 TO 1516 1591 TO 1594 1661 1664 1779 1782 TRAP JT -20 -20 -15 -15
1519 TO 1522 1597 TO 1600 1665 1668 1783 1786 TRAP JT -15 -15 -10 -10
1525 TO 1528 1603 TO 1606 1669 1672 1787 1790 TRAP JT -10 -10 -5 -5
1537 TO 1540 1615 TO 1618 1633 1636 1795 1798 TRAP JT -5 -5 0 0
LOAD 4 LOADTYPE Fluids  TITLE WATER PRESSURE2
ELEMENT LOAD
1678 1696 1747 1756 PR -25
1512 1551 1554 1583 1711 1727 TRAP JT -25 -25 -20 -20
1518 1555 1558 1589 1714 1730 TRAP JT -20 -20 -15 -15
1524 1559 1562 1595 1717 1733 TRAP JT -15 -15 -10 -10
1530 1563 1566 1601 1720 1736 TRAP JT -10 -10 -5 -5
1542 1571 1574 1613 1726 1742 TRAP JT -5 -5 0 0
1775 1778 TRAP JT 25 25 20 20
1779 1782 TRAP JT 20 20 15 15
1783 1786 TRAP JT 15 15 10 10
1787 1790 TRAP JT 10 10 5 5
1795 1798 TRAP JT 5 5 0 0
LOAD 5 LOADTYPE Soil  TITLE SOIL PRESSURE
ELEMENT LOAD
1507 TO 1510 1512 1543 1547 1551 1554 1575 1579 1583 1585 TO 1588 1641 1645 -
1649 1653 1657 1660 1709 TO 1711 1727 TO 1729 TRAP JT 18 18 13.5 13.5
1513 TO 1516 1518 1544 1548 1555 1558 1576 1580 1589 1591 TO 1594 1642 1646 -
1650 1654 1661 1664 1712 TO 1714 1730 TO 1732 TRAP JT 13.5 13.5 9 9
1519 TO 1522 1524 1545 1549 1559 1562 1577 1581 1595 1597 TO 1600 1643 1647 -
1651 1655 1665 1668 1715 TO 1717 1733 TO 1735 TRAP JT 9 9 4.5 4.5
1525 TO 1528 1530 1546 1550 1563 1566 1578 1582 1601 1603 TO 1606 1644 1648 -
1652 1656 1669 1672 1718 TO 1720 1736 TO 1738 TRAP JT 4.5 4.5 0 0
1507 TO 1510 1512 TO 1516 1518 TO 1522 1524 TO 1528 1530 1543 TO 1551 1554 -
1555 1558 1559 1562 1563 1566 1575 TO 1583 1585 TO 1589 1591 TO 1595 1597 -
1598 TO 1601 1603 TO 1606 1641 TO 1657 1660 1661 1664 1665 1668 1669 1672 -
1709 TO 1720 1727 TO 1738 PR 5
******FACTORED LOAD COMBINATIONS*****
LOAD COMB 101 1.4*(D+F1)
```

```
1 1.4 3 1.4
LOAD COMB 102 1.4*(D+F2)
1 1.4 4 1.4
LOAD COMB 103 1.4*(D++F1+F2)
1 1.4 3 1.4 4 1.4
LOAD COMB 104 1.2*(D+F1)+1.6*(L+H)
1 1.2 3 1.2 2 1.6 4 1.6
LOAD COMB 105 1.2*(D+F2)+1.6*(L+H)
1 1.2 4 1.2 2 1.6 4 1.6
LOAD COMB 106 1.2*(D+F1+F2)+1.6*(L+H)
1 1.2 3 1.2 4 1.2 2 1.6 4 1.6
LOAD COMB 107 1.4*(D+H)
1 1.4 4 1.4
******UNFACTORED LOAD COMBINATIONS*****
LOAD COMB 201 1.0*(D+F1)
1 1.0 3 1.0
LOAD COMB 202 1.0*(D+F2)
1 1.0 4 1.0
LOAD COMB 203 1.0*(D++F1+F2)
1 1.0 3 1.0 4 1.0
LOAD COMB 204 1.0*(D+F1)+1.0*(L+H)
1 1.0 3 1.0 2 1.0 4 1.0
LOAD COMB 205 1.0*(D+F2)+1.0*(L+H)
1 1.0 4 1.0 2 1.0 4 1.0
LOAD COMB 206 1.0*(D+F1+F2)+1.0*(L+H)
1 1.0 3 1.0 4 1.0 2 1.0 4 1.0
LOAD COMB 207 1.0*(D+H)
1 1.0 4 1.0
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 4 STAAD OUTPUT FOR SUMP TANK

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    OCT 15, 2020                *
*           Time=     10: 2:47                   *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
  1. STAAD SPACE
INPUT FILE: D:\MAADEN\Sewage Treatment Plant\Sump Tank\STAAD\SUMP TANK.STD
  2. START JOB INFORMATION
  3. ENGINEER DATE 08-OCT-20
  4. ENGINEER NAME BJB
  5. CHECKER NAME SJM
  6. END JOB INFORMATION
  7. INPUT WIDTH 79
  8. UNIT METER KN
  9. JOINT COORDINATES
 10. 1 0 0 0; 2 3.3 0 0; 3 0 0 1.2; 4 3.3 0 1.2; 5 0 -2.1 0; 6 3.3 -2.1 0
 11. 7 0 -2.1 1.2; 8 3.3 -2.1 1.2; 13 0 -2 0; 14 3.3 -2 0; 15 0 -2 1.2
 12. 16 3.3 -2 1.2; 17 0.1 0 0; 18 0.1 0 1.2; 19 0.1 -2.1 0; 20 0.1 -2.1 1.2
 13. 23 0.1 -2 0; 24 0.1 -2 1.2; 25 3.2 0 0; 26 3.2 0 1.2; 27 3.2 -2.1 0
 14. 28 3.2 -2.1 1.2; 31 3.2 -2 0; 32 3.2 -2 1.2; 33 0 0 0.1; 34 3.3 0 0.1
 15. 35 0 -2.1 0.1; 36 3.3 -2.1 0.1; 39 0 -2 0.1; 40 3.3 -2 0.1; 41 0 0 1.1
 16. 42 3.3 0 1.1; 43 0 -2.1 1.1; 44 3.3 -2.1 1.1; 47 0 -2 1.1; 48 3.3 -2 1.1
 17. 50 0.1 -2.1 0.1; 54 3.2 -2.1 0.1; 58 0.1 -2.1 1.1; 62 3.2 -2.1 1.1
 18. 943 0 0.5 0; 944 3.3 0.5 0; 945 0 0.5 1.2; 946 3.3 0.5 1.2; 947 0.1 0.5 0
 19. 948 0.1 0.5 1.2; 949 3.2 0.5 0; 950 3.2 0.5 1.2; 951 0 0.5 0.1
 20. 952 3.3 0.5 0.1; 953 0 0.5 1.1; 954 3.3 0.5 1.1; 1029 3.2 -2.1 0.6
 21. 1030 3.3 -2.1 0.6; 1035 3.3 -2 0.6; 1038 0.1 -2.1 0.6; 1039 0 -2.1 0.6
 22. 1044 0 -2 0.6; 1047 0.6 -2.1 1.2; 1048 0.6 -2.1 1.1; 1049 1.1 -2.1 1.2
 23. 1050 1.1 -2.1 1.1; 1051 1.6 -2.1 1.2; 1052 1.6 -2.1 1.1; 1053 2.1 -2.1 1.2
 24. 1054 2.1 -2.1 1.1; 1055 2.75 -2.1 1.2; 1056 2.75 -2.1 1.1; 1057 2.75 -2 1.2
 25. 1058 2.1 -2 1.2; 1059 1.6 -2 1.2; 1060 1.1 -2 1.2; 1061 0.6 -2 1.2
 26. 1062 0.6 -2.1 0.1; 1063 0.6 -2.1 0; 1064 1.1 -2.1 0.1; 1065 1.1 -2.1 0
 27. 1066 1.6 -2.1 0.1; 1067 1.6 -2.1 0; 1068 2.1 -2.1 0.1; 1069 2.1 -2.1 0
 28. 1070 2.75 -2.1 0.1; 1071 2.75 -2.1 0; 1072 0.6 -2 0; 1073 1.1 -2 0
 29. 1074 1.6 -2 0; 1075 2.1 -2 0; 1076 2.75 -2 0; 1077 0.6 -1.5 0; 1078 0.1 -1.5 0
 30. 1079 1.1 -1.5 0; 1080 1.6 -1.5 0; 1081 2.1 -1.5 0; 1082 2.75 -1.5 0
 31. 1083 3.2 -1.5 0; 1084 0.6 -1 0; 1085 0.0999999 -1 0; 1086 1.1 -1 0
 32. 1087 1.6 -1 0; 1088 2.1 -1 0; 1089 2.75 -1 0; 1090 3.2 -1 0; 1091 0.6 -0.5 0
 33. 1092 0.0999999 -0.5 0; 1093 1.1 -0.5 0; 1094 1.6 -0.5 0; 1095 2.1 -0.5 0
 34. 1096 2.75 -0.5 0; 1097 3.2 -0.5 0; 1098 0.6 0 0; 1099 1.1 0 0; 1100 1.6 0 0
 35. 1101 2.1 0 0; 1102 2.75 0 0; 1108 0.6 0.5 0; 1109 1.1 0.5 0; 1110 1.6 0.5 0
 36. 1111 2.1 0.5 0; 1112 2.75 0.5 0; 1113 3.3 -1.5 0; 1114 3.3 -1 0
 37. 1115 3.3 -0.5 0; 1116 3.3 -1.5 0.1; 1117 3.3 -1 0.1; 1118 3.3 -0.5 0.1
 38. 1119 3.3 -1.5 0.6; 1122 3.3 -1.5 1.1; 1123 3.3 -1 0.6; 1126 3.3 -1 1.1
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39. 1127 3.3 -0.5 0.6; 1130 3.3 -0.5 1.1; 1131 3.3 0 0.6; 1137 3.3 0.5 0.6
40. 1140 3.3 -1.5 1.2; 1141 3.3 -1 1.2; 1142 3.3 -0.5 1.2; 1143 3.2 -1.5 1.2
41. 1144 3.2 -1 1.2; 1145 3.2 -0.5 1.2; 1146 2.75 -1.5 1.2; 1147 2.1 -1.5 1.2
42. 1148 1.6 -1.5 1.2; 1149 1.1 -1.5 1.2; 1150 0.6 -1.5 1.2
43. 1151 0.0999999 -1.5 1.2; 1152 2.75 -1 1.2; 1153 2.1 -1 1.2; 1154 1.6 -1 1.2
44. 1155 1.1 -1 1.2; 1156 0.6 -1 1.2; 1157 0.0999999 -1 1.2; 1158 2.75 -0.5 1.2
45. 1159 2.1 -0.5 1.2; 1160 1.6 -0.5 1.2; 1161 1.1 -0.5 1.2; 1162 0.6 -0.5 1.2
46. 1163 0.1 -0.5 1.2; 1164 2.75 0 1.2; 1165 2.1 0 1.2; 1166 1.6 0 1.2
47. 1167 1.1 0 1.2; 1168 0.6 0 1.2; 1174 2.75 0.5 1.2; 1175 2.1 0.5 1.2
48. 1176 1.6 0.5 1.2; 1177 1.1 0.5 1.2; 1178 0.6 0.5 1.2; 1188 0 0.5 0.6
49. 1193 0 0 0.6; 1199 0 -1.5 0; 1200 0 -1 0; 1201 0 -0.5 0; 1202 0 -1.5 0.1
50. 1203 0 -1 0.1; 1204 0 -0.5 0.1; 1205 0 -1.5 1.2; 1206 0 -1 1.2
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52. 1211 0 -1.5 0.6; 1214 0 -1 0.6; 1217 0 -0.5 0.6; 1220 0.6 -2.1 0.6
53. 1221 1.1 -2.1 0.6; 1222 1.6 -2.1 0.6; 1223 2.1 -2.1 0.6; 1224 2.75 -2.1 0.6
54. 1235 2.20333 -2.1 1.2; 1236 2.20333 -2.1 1.1; 1237 2.30667 -2.1 1.2
55. 1238 2.30667 -2.1 1.1; 1239 2.30667 -2 1.2; 1240 2.20333 -2 1.2
56. 1241 2.20333 -2.1 0.1; 1242 2.20333 -2.1 0; 1243 2.30667 -2.1 0.1
57. 1244 2.30667 -2.1 0; 1245 2.20333 -2 0; 1246 2.30667 -2 0; 1247 2.20333 -1.5 0
58. 1248 2.30667 -1.5 0; 1249 2.20333 -1 0; 1250 2.30667 -1 0; 1251 2.20333 -0.5 0
59. 1252 2.30667 -0.5 0; 1253 2.20333 0 0; 1254 2.30667 0 0; 1257 2.20333 0.5 0
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61. 1261 2.30667 -1 1.2; 1262 2.20333 -1 1.2; 1263 2.30667 -0.5 1.2
62. 1264 2.20333 -0.5 1.2; 1265 2.30667 0 1.2; 1266 2.20333 0 1.2
63. 1269 2.30667 0.5 1.2; 1270 2.20333 0.5 1.2; 1271 2.20333 -2.1 0.6
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68. 1291 2.20333 0 1.1; 1293 2.20333 0.5 1.1; 1294 2.20333 -1.5 0.6
69. 1297 2.20333 -1 0.6; 1300 2.20333 -0.5 0.6; 1303 2.20333 0 0.6
70. 1309 2.20333 0.5 0.6
71. MEMBER INCIDENCES
72. 1799 943 947; 1800 947 1108; 1801 1108 1109; 1802 1109 1110; 1803 1110 1111
73. 1804 1111 1257; 1805 1257 1258; 1806 1258 1112; 1807 1112 949; 1808 949 944
74. 1809 945 948; 1810 948 1178; 1811 1178 1177; 1812 1177 1176; 1813 1176 1175
75. 1814 1175 1270; 1815 1270 1269; 1816 1269 1174; 1817 1174 950; 1818 950 946
76. 1819 943 951; 1820 951 1188; 1821 1188 953; 1822 953 945; 1823 944 952
77. 1824 952 1137; 1825 1137 954; 1826 954 946; 1827 1257 1287; 1828 1287 1309
78. 1829 1309 1293; 1830 1293 1270
79. ELEMENT INCIDENCES SHELL
80. 69 6 36 40 14; 70 27 6 14 31; 71 44 8 16 48; 72 8 28 32 16; 73 20 7 15 24
81. 74 7 43 47 15; 75 5 19 23 13; 76 35 5 13 39; 977 54 36 6 27; 978 35 50 19 5
82. 979 7 20 58 43; 980 28 8 44 62; 1181 2 34 952 944; 1182 25 2 944 949
83. 1183 42 4 946 954; 1184 4 26 950 946; 1185 18 3 945 948; 1186 3 41 953 945
84. 1187 1 17 947 943; 1188 33 1 943 951; 1467 54 1029 1030 36
85. 1470 1029 62 44 1030; 1471 36 1030 1035 40; 1474 1030 44 48 1035
86. 1475 58 1038 1039 43; 1478 1038 50 35 1039; 1479 43 1039 1044 47
87. 1482 1039 35 39 1044; 1483 20 1047 1048 58; 1484 1047 1049 1050 1048
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89. 1489 28 1055 1057 32; 1491 1053 1051 1059 1058; 1492 1051 1049 1060 1059
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93. 1503 1065 1067 1074 1073; 1504 1067 1069 1075 1074; 1506 1071 27 31 1076
94. 1507 23 1072 1077 1078; 1508 1072 1073 1079 1077; 1509 1073 1074 1080 1079
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95. 1510 1074 1075 1081 1080; 1512 1076 31 1083 1082; 1513 1078 1077 1084 1085
96. 1514 1077 1079 1086 1084; 1515 1079 1080 1087 1086; 1516 1080 1081 1088 1087
97. 1518 1082 1083 1090 1089; 1519 1085 1084 1091 1092; 1520 1084 1086 1093 1091
98. 1521 1086 1087 1094 1093; 1522 1087 1088 1095 1094; 1524 1089 1090 1097 1096
99. 1525 1092 1091 1098 17; 1526 1091 1093 1099 1098; 1527 1093 1094 1100 1099
100. 1528 1094 1095 1101 1100; 1530 1096 1097 25 1102; 1537 17 1098 1108 947
101. 1538 1098 1099 1109 1108; 1539 1099 1100 1110 1109; 1540 1100 1101 1111 1110
102. 1542 1102 25 949 1112; 1543 31 14 1113 1083; 1544 1083 1113 1114 1090
103. 1545 1090 1114 1115 1097; 1546 1097 1115 2 25; 1547 14 40 1116 1113
104. 1548 1113 1116 1117 1114; 1549 1114 1117 1118 1115; 1550 1115 1118 34 2
105. 1551 40 1035 1119 1116; 1554 1035 48 1122 1119; 1555 1116 1119 1123 1117
106. 1558 1119 1122 1126 1123; 1559 1117 1123 1127 1118; 1562 1123 1126 1130 1127
107. 1563 1118 1127 1131 34; 1566 1127 1130 42 1131; 1571 34 1131 1137 952
108. 1574 1131 42 954 1137; 1575 48 16 1140 1122; 1576 1122 1140 1141 1126
109. 1577 1126 1141 1142 1130; 1578 1130 1142 4 42; 1579 16 32 1143 1140
110. 1580 1140 1143 1144 1141; 1581 1141 1144 1145 1142; 1582 1142 1145 26 4
111. 1583 32 1057 1146 1143; 1585 1058 1059 1148 1147; 1586 1059 1060 1149 1148
112. 1587 1060 1061 1150 1149; 1588 1061 24 1151 1150; 1589 1143 1146 1152 1144
113. 1591 1147 1148 1154 1153; 1592 1148 1149 1155 1154; 1593 1149 1150 1156 1155
114. 1594 1150 1151 1157 1156; 1595 1144 1152 1158 1145; 1597 1153 1154 1160 1159
115. 1598 1154 1155 1161 1160; 1599 1155 1156 1162 1161; 1600 1156 1157 1163 1162
116. 1601 1145 1158 1164 26; 1603 1159 1160 1166 1165; 1604 1160 1161 1167 1166
117. 1605 1161 1162 1168 1167; 1606 1162 1163 18 1168; 1613 26 1164 1174 950
118. 1615 1165 1166 1176 1175; 1616 1166 1167 1177 1176; 1617 1167 1168 1178 1177
119. 1618 1168 18 948 1178; 1633 41 1193 1188 953; 1636 1193 33 951 1188
120. 1641 13 23 1078 1199; 1642 1199 1078 1085 1200; 1643 1200 1085 1092 1201
121. 1644 1201 1092 17 1; 1645 39 13 1199 1202; 1646 1202 1199 1200 1203
122. 1647 1203 1200 1201 1204; 1648 1204 1201 1 33; 1649 24 15 1205 1151
123. 1650 1151 1205 1206 1157; 1651 1157 1206 1207 1163; 1652 1163 1207 3 18
124. 1653 15 47 1208 1205; 1654 1205 1208 1209 1206; 1655 1206 1209 1210 1207
125. 1656 1207 1210 41 3; 1657 47 1044 1211 1208; 1660 1044 39 1202 1211
126. 1661 1208 1211 1214 1209; 1664 1211 1202 1203 1214; 1665 1209 1214 1217 1210
127. 1668 1214 1203 1204 1217; 1669 1210 1217 1193 41; 1672 1217 1204 33 1193
128. 1673 58 1048 1220 1038; 1674 1048 1050 1221 1220; 1675 1050 1052 1222 1221
129. 1676 1052 1054 1223 1222; 1678 1056 62 1029 1224; 1691 1038 1220 1062 50
130. 1692 1220 1221 1064 1062; 1693 1221 1222 1066 1064; 1694 1222 1223 1068 1066
131. 1696 1224 1029 54 1070; 1697 1053 1235 1236 1054; 1698 1235 1237 1238 1236
132. 1699 1237 1055 1056 1238; 1700 1055 1237 1239 1057; 1701 1237 1235 1240 1239
133. 1702 1235 1053 1058 1240; 1703 1068 1241 1242 1069; 1704 1241 1243 1244 1242
134. 1705 1243 1070 1071 1244; 1706 1069 1242 1245 1075; 1707 1242 1244 1246 1245
135. 1708 1244 1071 1076 1246; 1709 1075 1245 1247 1081; 1710 1245 1246 1248 1247
136. 1711 1246 1076 1082 1248; 1712 1081 1247 1249 1088; 1713 1247 1248 1250 1249
137. 1714 1248 1082 1089 1250; 1715 1088 1249 1251 1095; 1716 1249 1250 1252 1251
138. 1717 1250 1089 1096 1252; 1718 1095 1251 1253 1101; 1719 1251 1252 1254 1253
139. 1720 1252 1096 1102 1254; 1724 1101 1253 1257 1111; 1725 1253 1254 1258 1257
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142. 1732 1260 1147 1153 1262; 1733 1152 1261 1263 1158; 1734 1261 1262 1264 1263
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144. 1738 1264 1159 1165 1266; 1742 1164 1265 1269 1174; 1743 1265 1266 1270 1269
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147. 1756 1272 1224 1070 1243; 1757 1242 1241 1278 1245; 1758 1236 1235 1240 1277
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150. 1768 1253 1285 1287 1257; 1769 1277 1240 1260 1288; 1770 1288 1260 1262 1289

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151. 1771 1289 1262 1264 1290; 1772 1290 1264 1266 1291; 1774 1291 1266 1270 1293
152. 1775 1278 1279 1294 1282; 1778 1279 1277 1288 1294; 1779 1282 1294 1297 1283
153. 1782 1294 1288 1289 1297; 1783 1283 1297 1300 1284; 1786 1297 1289 1290 1300
154. 1787 1284 1300 1303 1285; 1790 1300 1290 1291 1303; 1795 1285 1303 1309 1287
155. 1798 1303 1291 1293 1309
156. START GROUP DEFINITION
157. ELEMENT
158. _WALL 1507 TO 1510 1512 TO 1516 1518 TO 1522 1524 TO 1528 1530 1537 TO 1540 -
159. 1542 1551 1554 1555 1558 1559 1562 1563 1566 1571 1574 1583 1585 TO 1589 -
160. 1591 TO 1595 1597 TO 1601 1603 TO 1606 1613 1615 TO 1618 1633 1636 1657 1660 -
161. 1661 1664 1665 1668 1669 1672 1711 1714 1717 1720 1726 1727 1730 1733 1736 -
162. 1742
163. _BOT_SLAB 1673 TO 1676 1678 1691 TO 1694 1696 1747 1756
164. _BUFFER_WALL 1775 1778 1779 1782 1783 1786 1787 1790 1795 1798
165. MEMBER
166. FLOOR
167. _GRATING 1799 TO 1830
168. END GROUP DEFINITION
169. ELEMENT PROPERTY
170. 69 TO 76 1181 TO 1188 1471 1474 1479 1482 1489 1491 TO 1494 1501 TO 1504 1506 -
171. 1507 TO 1510 1512 TO 1516 1518 TO 1522 1524 TO 1528 1530 1537 TO 1540 1542 -
172. 1543 TO 1551 1554 1555 1558 1559 1562 1563 1566 1571 1574 TO 1583 -
173. 1585 TO 1589 1591 TO 1595 1597 TO 1601 1603 TO 1606 1613 1615 TO 1618 1633 -
174. 1636 1641 TO 1657 1660 1661 1664 1665 1668 1669 1672 1700 TO 1702 -
175. 1706 TO 1720 1724 TO 1738 1742 TO 1744 THICKNESS 0.2
176. 977 TO 980 1467 1470 1475 1478 1483 TO 1486 1488 1495 TO 1498 1500 -
177. 1673 TO 1676 1678 1691 TO 1694 1696 TO 1699 1703 TO 1705 1745 TO 1747 1754 -
178. 1755 TO 1756 THICKNESS 0.2
179. 1757 TO 1759 1762 TO 1766 1768 TO 1772 1774 1775 1778 1779 1782 1783 1786 -
180. 1787 1790 1795 1798 THICKNESS 0.2
181. DEFINE MATERIAL START
182. ISOTROPIC CONCRETE
183. E 2.17185E+007
184. POISSON 0.17
185. DENSITY 25
186. ALPHA 1E-005
187. DAMP 0.05
188. TYPE CONCRETE
189. STRENGTH FCU 35000
190. ISOTROPIC DENSITYZERO
191. E 2.17185E+007
192. POISSON 0.17
193. DENSITY 0
194. ALPHA 1E-005
195. DAMP 0.05
196. TYPE CONCRETE
197. STRENGTH FCU 0
198. END DEFINE MATERIAL
199. MEMBER PROPERTY AMERICAN
200. 1799 TO 1830 PRIS YD 0.05 ZD 0.05
201. CONSTANTS
202. MATERIAL CONCRETE ALL
203. SUPPORTS
204. 977 TO 980 1467 1470 1475 1478 1483 TO 1486 1488 1495 TO 1498 1500 -
205. 1673 TO 1676 1678 1691 TO 1694 1696 TO 1699 1703 TO 1705 1745 TO 1747 1754 -
206. 1755 TO 1756 PLATE MAT DIRECT Y SUBGRADE 102960 PRINT
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JOINT	INFLUENCE AREA
5	0.002500
6	0.002500
7	0.002500
8	0.002500
19	0.015000
20	0.015000
27	0.013750
28	0.013750
35	0.015000
36	0.015000
43	0.015000
44	0.015000
50	0.090000
54	0.082500
58	0.090000
62	0.082500
1029	0.137500
1030	0.025000
1038	0.150000
1039	0.025000
1047	0.025000
1048	0.150000
1049	0.025000
1050	0.150000
1051	0.025000
1052	0.150000
1053	0.015083
1054	0.090500
1055	0.022333
1056	0.133999
1062	0.150000
1063	0.025000
1064	0.150000
1065	0.025000
1066	0.150000
1067	0.025000
1068	0.090500
1069	0.015083
1070	0.133999
1071	0.022333
1220	0.250000
1221	0.250000
1222	0.250000
1223	0.150833
1224	0.223332
1235	0.005167
1236	0.031000
1237	0.013667
1238	0.082000
1241	0.031000
1242	0.005167
1243	0.082000

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      1244          0.013667
      1271          0.051667
      1272          0.136667
     TOTAL          3.959999
  207. LOAD 1 LOADTYPE DEAD  TITLE DEAD LOAD
  208. SELFWEIGHT Y -1
  209. ELEMENT LOAD
  210. 1673 TO 1676 1678 1691 TO 1694 1696 1747 1756 PR -1.2
  211. FLOOR LOAD
  212. _GRATING FLOAD -0.05 GY
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4 Note 6.

  213. LOAD 2 LOADTYPE LIVE  TITLE LIVE LOAD
  214. FLOOR LOAD
  215. _GRATING FLOAD -5 GY
  216. LOAD 3 LOADTYPE FLUIDS  TITLE WATER PRESSURE1
  217. ELEMENT LOAD
  218. 1673 TO 1676 1691 TO 1694 PR -25
  219. 1507 TO 1510 1585 TO 1588 1657 1660 1775 1778 TRAP JT -25 -25 -20 -20
  220. 1513 TO 1516 1591 TO 1594 1661 1664 1779 1782 TRAP JT -20 -20 -15 -15
  221. 1519 TO 1522 1597 TO 1600 1665 1668 1783 1786 TRAP JT -15 -15 -10 -10
  222. 1525 TO 1528 1603 TO 1606 1669 1672 1787 1790 TRAP JT -10 -10 -5 -5
  223. 1537 TO 1540 1615 TO 1618 1633 1636 1795 1798 TRAP JT -5 -5 0 0
  224. LOAD 4 LOADTYPE FLUIDS  TITLE WATER PRESSURE2
  225. ELEMENT LOAD
  226. 1678 1696 1747 1756 PR -25
  227. 1512 1551 1554 1583 1711 1727 TRAP JT -25 -25 -20 -20
  228. 1518 1555 1558 1589 1714 1730 TRAP JT -20 -20 -15 -15
  229. 1524 1559 1562 1595 1717 1733 TRAP JT -15 -15 -10 -10
  230. 1530 1563 1566 1601 1720 1736 TRAP JT -10 -10 -5 -5
  231. 1542 1571 1574 1613 1726 1742 TRAP JT -5 -5 0 0
  232. 1775 1778 TRAP JT 25 25 20 20
  233. 1779 1782 TRAP JT 20 20 15 15
  234. 1783 1786 TRAP JT 15 15 10 10
  235. 1787 1790 TRAP JT 10 10 5 5
  236. 1795 1798 TRAP JT 5 5 0 0
  237. LOAD 5 LOADTYPE SOIL  TITLE SOIL PRESSURE
  238. ELEMENT LOAD
  239. 1507 TO 1510 1512 1543 1547 1551 1554 1575 1579 1583 1585 TO 1588 1641 1645 -
  240. 1649 1653 1657 1660 1709 TO 1711 1727 TO 1729 TRAP JT 18 18 13.5 13.5
  241. 1513 TO 1516 1518 1544 1548 1555 1558 1576 1580 1589 1591 TO 1594 1642 1646 -
  242. 1650 1654 1661 1664 1712 TO 1714 1730 TO 1732 TRAP JT 13.5 13.5 9 9
  243. 1519 TO 1522 1524 1545 1549 1559 1562 1577 1581 1595 1597 TO 1600 1643 1647 -
  244. 1651 1655 1665 1668 1715 TO 1717 1733 TO 1735 TRAP JT 9 9 4.5 4.5
  245. 1525 TO 1528 1530 1546 1550 1563 1566 1578 1582 1601 1603 TO 1606 1644 1648 -
  246. 1652 1656 1669 1672 1718 TO 1720 1736 TO 1738 TRAP JT 4.5 4.5 0 0
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247. 1507 TO 1510 1512 TO 1516 1518 TO 1522 1524 TO 1528 1530 1543 TO 1551 1554 -
248. 1555 1558 1559 1562 1563 1566 1575 TO 1583 1585 TO 1589 1591 TO 1595 1597 -
249. 1598 TO 1601 1603 TO 1606 1641 TO 1657 1660 1661 1664 1665 1668 1669 1672 -
250. 1709 TO 1720 1727 TO 1738 PR 5
251. ******FACTORED LOAD COMBINATIONS*****
252. LOAD COMB 101 1.4*(D+F1)
253. 1 1.4 3 1.4
254. LOAD COMB 102 1.4*(D+F2)
255. 1 1.4 4 1.4
256. LOAD COMB 103 1.4*(D++F1+F2)
257. 1 1.4 3 1.4 4 1.4
258. LOAD COMB 104 1.2*(D+F1)+1.6*(L+H)
259. 1 1.2 3 1.2 2 1.6 4 1.6
260. LOAD COMB 105 1.2*(D+F2)+1.6*(L+H)
261. 1 1.2 4 1.2 2 1.6 4 1.6
262. LOAD COMB 106 1.2*(D+F1+F2)+1.6*(L+H)
263. 1 1.2 3 1.2 4 1.2 2 1.6 4 1.6
264. LOAD COMB 107 1.4*(D+H)
265. 1 1.4 4 1.4
266. ******UNFACTORED LOAD COMBINATIONS*****
267. LOAD COMB 201 1.0*(D+F1)
268. 1 1.0 3 1.0
269. LOAD COMB 202 1.0*(D+F2)
270. 1 1.0 4 1.0
271. LOAD COMB 203 1.0*(D++F1+F2)
272. 1 1.0 3 1.0 4 1.0
273. LOAD COMB 204 1.0*(D+F1)+1.0*(L+H)
274. 1 1.0 3 1.0 2 1.0 4 1.0
275. LOAD COMB 205 1.0*(D+F2)+1.0*(L+H)
276. 1 1.0 4 1.0 2 1.0 4 1.0
277. LOAD COMB 206 1.0*(D+F1+F2)+1.0*(L+H)
278. 1 1.0 3 1.0 4 1.0 2 1.0 4 1.0
279. LOAD COMB 207 1.0*(D+H)
280. 1 1.0 4 1.0
281. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**241**	**NUMBER OF MEMBERS**	**32**
NUMBER OF PLATES	**232**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**55**

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 1281

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282. FINISH
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********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 15,2020 TIME= 10: 2:49 ****